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This is a corrected copy of the paper cited above. The text is retyped from the printed paper. It keeps the paper's wording and the numbering of its results, equations and footnotes, and a marker such as [p. N] shows where page N of the printed paper begins.

The corrections come from a formalization of the Graph Minors series in the Lean proof assistant, github.com/mahdi-jfri/GraphMinor, in which the results of this paper are proved. Where the formalization found that a statement or a step of a proof is wrong as printed, this copy corrects it in place. Where the paper leaves a proof or a step to the reader, this copy adds it, also in blue. Text ~~struck through in red~~ is removed and text in blue is inserted. Each change has a note, marked F1, F2, ..., that gives the reason. The file corrections/GM12.md of the formalization describes each correction in full. Each note, and each item in the list below, begins with the kind of the change. A **misprint** is a wrong symbol, index or word. A **statement mistake** is a result or a claim of the text that is false as printed. A **degenerate case** is a statement that fails only in a trivial case, such as an empty graph or a parameter equal to 0. A **proof gap** is a step of a proof that does not follow as written. A **missing proof** is a proof that the paper omits, leaves to the reader or only sketches, and that this copy adds.

The corrections and added proofs, in the order of the pages:

1. **Missing proof.** Section 1, page 242: a proof of the observation on $a(c)$ and $ins(W)$ is added.
2. **Degenerate case.** Statement of (1.2), page 242: the hypothesis that Γ has at least two edges is added.
3. **Misprint.** Section 2, page 243: in clause (ii) of the definition of a trace, I_i becomes I_{i-1}.
4. **Missing proof.** Section 2, page 243: the proof that the edge e_{2i} of clause (iii) exists and is unique is added.
5. **Missing proof.** Proof of (4.2), page 249: the proof is added.
6. **Degenerate case.** Statement of (5.2), page 254: the hypothesis $\theta \geq 2$ is added.

7. **Proof gap.** Proof of (5.3), page 256: a note explains how to deal with infinitely many components of $F - \Sigma_2$.
8. **Proof gap.** Proof of (5.2), page 257: "immediately" is removed, and a note explains how (5.2) follows from (5.3).
9. **Degenerate case.** Statement of (6.1), page 258: the hypothesis $\theta \geq 4$ is added.
10. **Proof gap.** Proof of (6.1), page 258: a proof is inserted that the metric d does not depend on the radial drawing used to define it.
11. **Proof gap.** Proof of (6.1), page 258: a proof is inserted that no closed walk W^* of K of length $< 2\theta$ with $U(W^*) \subseteq \Delta$ has e_1 and $e_{\theta/2}$ in its inside.
12. **Proof gap.** Proof of (7.3), page 261: a note explains how to deal with loops at v that bound a disc.
13. **Proof gap.** Proof of (7.5), page 265: a note explains why Graph Minors XI (6.6) may be used here for order 2.
14. **Misprint.** Statement of (7.8), page 268: "if $c \in E(\Gamma')$" becomes "if $c \in E(\Gamma) - E(\Gamma')$".
15. **Statement mistake.** Statement of (7.11), page 271: the hypothesis that $\lambda \geq 2$ if $z \in E(\Gamma)$ is added.

[p. 240] # Graph Minors.
XII. Distance on a Surface

Neil Robertson*
Department of Mathematics, Ohio State University, 231 West 18th Avenue, Columbus, Ohio 43210
AND
P. D. Seymour
Bellcore, 445 South Street, Morristown, New Jersey 07960
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Let Γ be a graph drawn on a connected surface Σ which is not a sphere. It is "θ-representative" if every non-null-homotopic closed curve meets Γ at least θ times. Also, Γ defines a metric on Σ, discussed in an earlier paper. Our objective here is to study the effect on the metric and on the "representativeness" of making local changes in the drawing or in the surface. We also reformulate more compactly the main theorem of an earlier paper in terms of this metric. These are lemmas to be used later.

1. Introduction

In future papers of this series we shall prove a theorem about the structure of the graphs not containing a fixed graph as a minor. Our objective here is to establish certain lemmas towards that proof, concerning drawings of graphs in surfaces, and the associated metric, introduced in [5].

A surface is a compact 2-manifold, without boundary. Let Σ be a surface. A line in Σ is a subset homeomorphic to $[0,1]$, and its ends are defined in the natural way. An O-arc is a subset homeomorphic to a circle. If $\Delta \subseteq \Sigma$, $\bar{\Delta}$ denotes the closure of Δ, and $bd(\Delta)$ denotes $\bar{\Delta} \cap \overline{\Sigma - \Delta}$. A drawing in Σ is a pair (U, V), where $V \subseteq U \subseteq \Sigma$, U is closed, V is finite, $U - V$ has only finitely many arc-wise connected components, called *edges*, and for every edge e, either $\bar{e}$ is a line with set of ends $\bar{e} \cap V$, or $\bar{e}$ is an O-arc and $|\bar{e} \cap V| = 1$. The components of $\Sigma - U$ are called *regions*. The members of [p. 241] V are called vertices. If $\Gamma = (U, V)$ is a drawing, we write $U = U(\Gamma)$, $V = V(\Gamma)$, and $E(\Gamma)$, $R(\Gamma)$ are defined to be the sets of edges and regions of Γ, respectively. The sets $\{v\}$ $(v \in V(\Gamma))$ and the edges and regions of Γ are called the *atoms* of Γ; the atoms form a partition of Σ. The set of

*Part of this work was performed under a consulting agreement with Bellcore.

atoms is denoted by $A(\Gamma)$. If v is a vertex and e is an edge or region of a drawing Γ, we say v is *incident with* e (and vice versa) if v is contained in the closure of e. Thus $V(\Gamma)$, $E(\Gamma)$ and the incidence relation between them define a graph, and in this sense we may regard Γ as a graph, and use graph-theoretic terminology for drawings. If e is an edge of Γ and r is a region, we say e is *incident with* r (and vice versa) if e is included in the closure of r. We say Γ' is a *subdrawing* of a drawing Γ, and write $\Gamma' \subseteq \Gamma$, if $V(\Gamma') \subseteq V(\Gamma)$ and $E(\Gamma') \subseteq E(\Gamma)$ (hence $U(\Gamma') \subseteq U(\Gamma)$). If Γ_1, Γ_2 are subdrawings of Γ, we define $\Gamma_1 \cup \Gamma_2$, $\Gamma_1 \cap \Gamma_2$ in the natural way. A drawing Γ in Σ is *2-cell* if Σ is connected and every region is an open disc (and hence Γ is not null). Let Σ be a surface, and let Γ be a drawing in Σ. A subset $Z \subseteq \Sigma$ is Γ-*normal* if $Z \cap U(\Gamma) \subseteq V(\Gamma)$. If Σ is connected and not a sphere, we say that Γ is θ-*representative* if $|F \cap V(\Gamma)| \geq \theta$ for every non-null-homotopic Γ-normal O-arc F in Σ. If $\Delta \subseteq \Sigma$ is closed and $bd(\Delta)$ is Γ-normal then $(U(\Gamma) \cap \Delta, V(\Gamma) \cap \Delta)$ is a drawing which we denote by $\Gamma \cap \Delta$.

A separation of a graph G is a pair (A, B) of subgraphs with $A \cup B = G$ and $E(A \cap B) = \emptyset$, and its order is $|V(A \cap B)|$. A tangle of order $\theta \geq 1$ in G is a set $\mathscr{T}$ of separations of G, each of order $< \theta$, such that

(i) for every separation (A, B) of G of order $< \theta$, $\mathscr{T}$ contains one of (A, B), (B, A)

(ii) if (A_1, B_1), (A_2, B_2), $(A_3, B_3) \in \mathscr{T}$ then $A_1 \cup A_2 \cup A_3 \neq G$

(iii) if $(A, B) \in \mathscr{T}$ then $V(A) \neq V(G)$.

We refer to these as the *first*, *second*, and *third tangle axioms*. Tangles were introduced in [4].

A tangle $\mathscr{T}$ in a drawing Γ of order θ is said to be *respectful* if Σ is connected and for every Γ-normal O-arc F in Σ with $|F \cap V(\Gamma)| < \theta$, there is a closed disc $\Delta \subseteq \Sigma$ with $bd(\Delta) = F$ such that

$$(\Gamma \cap \Delta, \Gamma \cap \overline{\Sigma - \Delta}) \in \mathscr{T}.$$

Then Δ is necessarily unique, and we write $\Delta = ins(F)$; the function ins is called the *inside function* of $\mathscr{T}$. (A drawing Γ may be a drawing in more than one surface, and if we wish to emphasize which surface we are dealing with, we say that $\mathscr{T}$ is *respectful towards* Σ.) If there is a respectful tangle of order θ in Γ and Σ is not a sphere, then clearly Γ is θ-representative.

Let Γ be a 2-cell drawing in Σ, and let K be a drawing in Σ. We say that K is a *radial drawing* of Γ if

[p. 242] (i) $U(K) \cap U(\Gamma) = V(\Gamma) \subseteq V(K)$

(ii) every region of Γ includes a unique vertex of K

(iii) K is bipartite, and $(V(\Gamma), V(K) - V(\Gamma))$ is a bipartition of it

(iv) for every $v \in V(\Gamma)$, the edges of $K \cup \Gamma$ incident with v belong alternately to Γ and to K (in their cyclic order around v).

Thus, loosely, K is obtained by adding a new vertex in each region of Γ and making it adjacent to every vertex of Γ incident with the region. If $E(\Gamma) \neq \emptyset$ and K is a radial drawing of Γ, it is not difficult to show that K is 2-cell, and every region of K includes a unique edge of Γ.

Let Γ be a 2-cell drawing in a connected surface Σ, with a respectful tangle of order θ, and let K be a radial drawing of Γ. For every circuit C of K of length $< 2\theta$, we define $ins(C) = ins(U(C))$. We shall often need the following result, a consequence of [5, Theorem (6.3)].

(1.1) *In these circumstances, let $E(\Gamma) \neq \emptyset$, and let (A, B) be a separation of Γ of order $< \theta$. Then $(A, B) \in \mathscr{T}$ if and only if for every edge e of A there is a circuit C of K with $V(C) \cap V(\Gamma) \subseteq V(A \cap B)$ and with $e \subseteq ins(C)$.*

If W is a closed walk of K, we define $K|W$ to be the subdrawing of K formed by the vertices and edges in W; and we write $U(W) = U(K|W)$, $V(W) = V(K|W)$, etc. If W has length $< 2\theta$ we define $ins(W)$ to be the union of $U(W)$ and $ins(C)$, taken over all circuits C of $K|W$. We recall that $A(K)$ is the set of atoms of K. If $a, b \in A(K)$, we define $d(a, b)$ by:

(i) if $a = b$ then $d(a, b) = 0$

(ii) if $a \neq b$ and $a, b \subseteq ins(W)$ for some closed walk W of K of length $< 2\theta$, then $d(a, b)$ is half the minimum length of such a walk

(iii) if (i) and (ii) do not apply then $d(a, b) = \theta$.

Now for each atom c of Γ there corresponds an atom a of K; namely, $a = c$ if $c \subseteq V(\Gamma)$; a is the region of K including c if c is an edge of Γ; and $a = \{v\}$, where v is the vertex of K in c, if c is a region of Γ. Let us write $a = a(c)$; then for $c_1, c_2 \in A(\Gamma)$ we define $d(c_1, c_2) = d(a(c_1), a(c_2))$. We shall frequently use the result of [5, Theorem (9.1)], that d defines a metric on $A(K)$ and hence on $A(\Gamma)$; and the easy observation that for a closed walk W in K of length $< 2\theta$, and for $c \in A(\Gamma)$, $a(c) \subseteq ins(W)$ if and only if $c \cap ins(W) \neq \emptyset$.

We prove this observation. Each circuit C of $K|W$ has length $< 2\theta$, and so $ins(C)$ is defined, and it is a closed disc with boundary $U(C)$. We use two facts. First, an edge of Γ misses $U(K)$, by (i). Second, let L be a subdrawing of K with a point in a region c of Γ, and let v be the vertex of K in c. Then $v \in V(L)$. For the point is v or lies in an edge of L, and this edge lies in c, since it misses $U(\Gamma)$ by (i). By (iii) it has an end not in $V(\Gamma)$, and this end is not in $U(\Gamma)$, by (i), and so it is in c, and it is v by (ii).

Now if $a(c) \subseteq ins(W)$, then $c \cap ins(W) \neq \emptyset$, since $a(c)$ and c are non-empty and one of them includes the other. Conversely, let $x \in c \cap ins(W)$.

If $c \subseteq V(\Gamma)$, then $a(c) = c = \{x\}$, and so $a(c) \subseteq ins(W)$. If c is an edge of Γ, then $x \notin U(W)$ by the first fact, and so $x \in ins(C)$ for some circuit C of $K|W$. The region $a(c)$ of K is connected, it misses $U(C)$, which is the boundary of $ins(C)$, and it meets $ins(C)$ at x. So $a(c) \subseteq ins(C) \subseteq ins(W)$. Finally, let c be a region of Γ, and let v be the vertex of K in c. If $x \in U(W)$, then $v \in V(W)$ by the second fact, applied to $K|W$, and so $a(c) = \{v\} \subseteq ins(W)$. Otherwise $x \in ins(C)$ for some circuit C of $K|W$. If $v \in V(C)$, then $a(c) \subseteq ins(W)$. If not, then c misses $U(C)$ by the second fact, applied to $K|C$. Since c is connected and meets $ins(C)$ at x, it follows that $c \subseteq ins(C)$, and so $a(c) \subseteq ins(W)$.[F1]

We shall also frequently need the following, from [5, Theorem (8.12)].

(1.2) *Let $\mathscr{T}$ be a respectful tangle of order $\theta \geq 2$ in a 2-cell drawing Γ with $|E(\Gamma)| \geq 2$ in a connected surface Σ, and let its metric be d. Let $c \in A(\Gamma)$; then there is an edge e of Γ with $d(c, e) = \theta$.*[F2]

[p. 243] 2. The Trace of a Closed Curve

Let Γ be a 2-cell drawing in a connected surface Σ, and let K be a radial drawing of Γ. Let $\phi\colon S^1 \to \Sigma$ be continuous. (S^1 is the unit circle $\{(x, y)\colon x^2 + y^2 = 1\}$.) If $\{x \in S^1\colon \phi(x) \in U(\Gamma)\}$ is finite, we call its cardinality the *length* of ϕ (with respect to Γ). We denote $\{\phi(x)\colon x \in S^1\}$ by $|\phi|$. Now let ϕ have finite length and let $|\phi|$ be Γ-normal. We define a *trace* of ϕ in K as follows. If ϕ has length 0, its only trace is the zero-edge walk with one vertex v, where $v \in V(K)$ is the vertex of K in the region of Γ including $|\phi|$. If ϕ has length $n > 0$, we proceed as follows. Let

$$S^1 = I_1 \cup \cdots \cup I_n \cup \{x_1, \ldots, x_n\},$$

where $x_1, \ldots, x_n \in S^1$ are distinct, $\{x_1, \ldots, x_n\} = \{x \in S^1\colon \phi(x) \in V(\Gamma)\}$, $I_1, \ldots, I_n$ are mutually disjoint, arc-wise connected open subsets of $S^1 - \{x_1, \ldots, x_n\}$, and for $1 \leq i \leq n$, $\bar{I}_i - I_i = \{x_i, x_{i+1}\}$ (where x_{n+1} means x_1). Let W be the closed walk $v_0, e_1, v_1, e_2, \ldots, e_{2n}, v_{2n} = v_0$ in K, defined by

(i) For $1 \leq i \leq n$, $v_{2i} = \phi(x_i)$, and $v_0 = v_{2n}$.

(ii) For $1 \leq i \leq n$, v_{2i-1} is the vertex of K in r_{2i-1}, where r_{2i-1} is the region of Γ including $\{\phi(x)\colon x \in \not{I}_i I_{i-1}\}$[F3].

[F1] **Missing proof.** The paper calls this an easy observation and gives no proof. The argument above is the one used in the formalization.

[F2] **Degenerate case.** As printed, (1.2) fails for the drawing in the sphere with two vertices and one edge e_0 joining them, and a respectful tangle of order 2 in it. For $c = e_0$ the only edge is e_0, and $d(e_0, e_0) = 0 \neq 2$. If $|E(\Gamma)| \geq 2$, then (1.2) holds.

[F3] **Misprint.** Here v_{2i-1} lies between $v_{2i-2} = \phi(x_{i-1})$ and $v_{2i} = \phi(x_i)$, so its region

(iii) For $1 \le i \le n$, e_{2i} is the edge of K joining v_{2i-1} and v_{2i} which enters v_{2i} "in the same way as ϕ." More precisely, choose $h \in I_{i-1}$; then for any $e \in E(K)$ joining v_{2i-1} and v_{2i}, we can construct a continuous ϕ_e consisting of the restriction of ϕ to the portion of I_{i-1} between h and x_i, concatenated with a curve tracing e from v_{2i} to v_{2i-1}. There is a unique edge e such that ϕ_e is homotopic in $r_{2i-1} \cup \{v_{2i}\}$ to a curve in r_{2i-1}, and we define e_{2i} to be this edge e. (We ~~omit~~give below the proof that e exists and is unique and does not depend on h.)

(iv) Similarly, for $1 \le i \le n$, e_{2i-1} is the edge of K joining v_{2i-2} and v_{2i-1} which leaves v_{2i-2} in the same way as ϕ.

Such a walk W is said to be a *trace* of ϕ.

We prove the assertion omitted in (iii). The assertion used in (iv) follows from it by running ϕ backwards. Let $v = v_{2i}$ and $r = r_{2i-1}$, and suppose that Γ has an edge. (If Γ has no edge, then by condition (iv) on page 242, K has no edge, and so a curve of positive length has no trace.) Then v is an end of an edge of Γ. For otherwise a small circle around v bounds a closed disc in the region of Γ around v, since this region is an open disc. These two closed discs have the same boundary and are different, since only the small one contains v, and so their union is open and closed in Σ. Since Σ is connected, they cover Σ, and so $U(\Gamma) = \{v\}$, a contradiction.

For an edge e of K joining v_{2i-1} and v, and for $h \in I_{i-1}$, let us say that e *enters* v *like* ϕ *from* h if ϕ_e, formed with this h, is homotopic in $r \cup \{v\}$ to a curve in r. Since ϕ maps I_{i-1} into r, adding to ϕ_e or removing from it an initial part of ϕ in r does not change this property. So if e enters v like ϕ from one h, then it does so from every $h \in I_{i-1}$.

First we show that some edge e enters v like ϕ from some h. Suppose first that at least two edges of Γ are incident with v, a loop counted twice. Then the edges of Γ at v divide a small closed disc around v into closed sectors, each a closed disc with v in its boundary, and the points of the small disc not in $U(\Gamma)$ are interior points of the sectors. Since ϕ maps the part of I_{i-1} near x_i into r, which misses $U(\Gamma)$, and this part is connected, ϕ maps it into the interior of one sector Q. By condition (iv) on page 242, the edges of Γ and of K at v alternate in the cyclic order around v, and so some edge e of K at v starts inside Q. Let P be the set formed by v and the interior points of Q near v. It is simply connected, since it contracts to v, and $P - \{v\}$ is path connected. Also $P - \{v\}$ misses $U(\Gamma)$ and contains points of $\phi(I_{i-1}) \subseteq r$, and so it lies in r. Since e is connected and misses

includes $\phi(I_{i-1})$, where I_0 means I_n. Clause (iii) uses I_{i-1} in the same way. With this indexing, (2.1) holds.

$U(\Gamma)$, and its points near v are in $P - \{v\}$, it lies in r, and so its other end is v_{2i-1}, the vertex of K in r. Choose $h \in I_{i-1}$ such that ϕ maps the portion of I_{i-1} between h and x_i into P, choose a point z of e such that e runs inside P from v to z, and join $\phi(h)$ to z by a path in $P - \{v\}$. Since P is simply connected, the restriction of ϕ to the portion of I_{i-1} between h and x_i, followed by a curve tracing e from v to z, is homotopic in P to this path. Hence ϕ_e is homotopic in $r \cup \{v\}$ to this path followed by a curve tracing e from z to v_{2i-1}, which is a curve in r. So e enters v like ϕ from h.

Now suppose that just one edge f of Γ is incident with v, and it is not a loop. By condition (iv) on page 242, exactly one edge e of K is incident with v. Remove from a small disc around v a short initial part of f. The rest of this disc is a simply connected set P, and $P - \{v\}$ is path connected and misses $U(\Gamma)$. It meets r, since v is in the closure of r, and so it lies in r. The points of e near v are in $P - \{v\}$, and so, as before, e lies in r and joins v and v_{2i-1}. Also ϕ maps the part of I_{i-1} near x_i into P, since this part is mapped near v and into r. So the same argument with this set P shows that e enters v like ϕ.

Next let e, e' be edges of K joining v_{2i-1} and v that enter v like ϕ, from h and from h' say. By the second paragraph, both enter v like ϕ from h. So ϕ_e and $\phi_{e'}$, formed with h, are homotopic in $r \cup \{v\}$ to curves in r with the same ends $\phi(h)$ and v_{2i-1}. These two curves are homotopic in r, since r is an open disc, and so ϕ_e and $\phi_{e'}$ are homotopic in $r \cup \{v\}$. Cancelling their common initial part, we find that the curves tracing e and e' from v to v_{2i-1} are homotopic in $r \cup \{v\}$. If $e \neq e'$, then $\bar{e} \cup \bar{e}'$ is an O-arc null-homotopic in $r \cup \{v\}$, and so by [5, Theorem (7.2)] it bounds a disc $\Delta \subseteq r \cup \{v\}$. Since $\Delta - \{v\} \subseteq r$ misses $U(\Gamma)$, no edge of Γ at v enters Δ, and so two edges of K are consecutive in the cyclic order around v, contrary to condition (iv) on page 242. So $e = e'$, which proves that e is unique and does not depend on h.[F4]

The main result of this section is the following.

(2.1) *Let Γ be a 2-cell drawing in a connected surface Σ, with a radial drawing K. Let $\mathscr{T}$ be a respectful tangle in Γ of order $\theta \geq 1$, let $\phi\colon S^1 \to \Sigma$ be continuous, of length $< \theta$, such that $|\phi|$ is Γ-normal, and let W be a trace of ϕ. Then $ins(W)$ is defined. Moreover, for any atom c of Γ, if there is an O-arc $F \subseteq |\phi|$ with $c \cap ins(F) \neq \emptyset$, then $c \cap ins(W) \neq \emptyset$.*

Proof. W has $2n < 2\theta$ edges, where ϕ has length n, and so $ins(W)$ is defined. Let $F \subseteq |\phi|$ be an O-arc with $c \cap ins(F) \neq \emptyset$; and let $\phi'\colon S^1 \to F$ be a homeomorphism. If W' is a trace of ϕ', then $K|W'$ is a subgraph of [p. 244]

[F4] **Missing proof.** The paper omits this proof, and we give it above.

$K \mid W$ as is easily seen; and hence if the result holds for ϕ' then it holds for ϕ. Thus we may assume, to simplify notation, that $\phi = \phi'$; in other words, it suffices to prove the result when ϕ is a simple closed curve. We prove this by induction on n, the length of ϕ. Let $W = v_0, e_1, v_1, \ldots, e_{2n}, v_{2n}$, as in the definition of "trace," and for $1 \leq i \leq n$ let r_{2i-1} be the region of Γ containing v_{2i-1}.

Case 1. $v_1, v_3, \ldots, v_{2n-1}$ are all distinct. Now either $n = 1$ and W is a walk v, e, u, e, v for some edge e of K with ends u, v, or $U(W)$ is a simple closed curve. In the first case the only atoms of Γ meeting $\mathrm{ins}(\phi)$ are the two corresponding to the atoms $\{u\}$, $\{v\}$ of K, and both of these are met by $\mathrm{ins}(W) = U(W)$ as required. We assume then that $U(W)$ is a simple closed curve; let $\Delta_2 = \mathrm{ins}(W)$, $\Delta_2' = \Sigma - \Delta_2$. Let $\Delta_1 = \mathrm{ins}(|\phi|)$, $\Delta_1' = \Sigma - \Delta_1$. Since W is a trace of ϕ, and so $U(W)$ can be obtained from $|\phi|$ by continuously deforming the portion of ϕ mapping into r_{2i-1}, for each i, it follows that

$$(\Gamma \cap \Delta_1, \Gamma \cap \Delta_1') = (\Gamma \cap \Delta_2, \Gamma \cap \Delta_2') \quad \text{or} \quad (\Gamma \cap \Delta_2', \Gamma \cap \Delta_2);$$

and the latter is impossible since $(\Gamma \cap \Delta_i, \Gamma \cap \Delta_i') \in \mathscr{T}$ $(i = 1, 2)$. Thus $\Gamma \cap \Delta_1 = \Gamma \cap \Delta_2$ and so Δ_1 and Δ_2 meet the same atoms of Γ; hence c meets $\Delta_2 = \mathrm{ins}(W)$, as required.

Case 2. $v_1, v_3, \ldots, v_{2n-1}$ are not all distinct. Thus there exists i with $1 \leq i \leq n$ such that $|\phi| \cap r_{2i-1}$ is not connected. Let F be a minimal line with $F \subseteq r_{2i-1}$ meeting two components of $|\phi| \cap r_{2i-1}$. Then no internal point of F belongs to $|\phi|$. Let F_1, F_2, $|\phi|$ be the three O-arcs included in $F \cup |\phi|$. Now $\mathrm{ins}(F_1) \cup \mathrm{ins}(F_2) \cup \mathrm{ins}(|\phi|) \neq \Sigma$, from the second tangle axiom, and so one of $\mathrm{ins}(F_1)$, $\mathrm{ins}(F_2)$, $\mathrm{ins}(|\phi|)$ is the union of the other two. It follows that c meets one of $\mathrm{ins}(F_1)$, $\mathrm{ins}(F_2)$, say, $\mathrm{ins}(F_1)$. Now ϕ_1 has length $< n$, where $\phi_1 \colon S^1 \to F_1$ is a homeomorphism; so by our inductive hypothesis, c meets $\mathrm{ins}(W_1)$, where W_1 is a trace of ϕ_1. But $K \mid W_1$ is a subdrawing of $K \mid W$; so c meets $\mathrm{ins}(W)$, as required. □

3. Disjoint Connecting Subgraphs

Our first objective is to reformulate the main theorem of [3] in terms of respectful tangles; we shall see that doing so yields a considerably simpler version of the theorem. Let Σ be a connected surface, and let $\Delta_1, \ldots, \Delta_t$ be mutually disjoint closed discs in Σ. Let Γ be a drawing in Σ such that $U(\Gamma) \cap \Delta_i = V(\Gamma) \cap \mathrm{bd}(\Delta_i)$ for $1 \leq i \leq t$; let

$$Z = \bigcup_{1 \leq i \leq t} V(\Gamma) \cap \mathrm{bd}(\Delta_i);$$

[p. 245] and let $(Z_j : 1 \le j \le p)$ be a partition of Z. We are concerned with the truth of the following statement.

There are mutually disjoint, connected subdrawings $H_1, \dots, H_p$ of Γ with $V(H_j) \cap Z = Z_j$ $(1 \le j \le p)$. (*)

One necessary condition for (*) to hold is the following.

Topological Feasibility Condition. There exist mutually disjoint closed discs $D_1, \dots, D_p \subseteq \Sigma$ such that

$$D_j \cap (\Delta_1 \cup \cdots \cup \Delta_t) = Z_j \quad (1 \le j \le p).$$

(We observe that this condition does not depend on Γ, but only on $\Delta_1, \dots, \Delta_t$ and $Z_1, \dots, Z_p$.) The following was shown in [3, Sections 7 and 8]. Let Σ^* be the closure of $\Sigma - (\Delta_1 \cup \cdots \cup \Delta_t)$.

(3.1) *With notation as above, suppose that either Σ is not a sphere or $t \ge 3$. Then there is an integer N, depending on Σ, t, $|Z|$ but not on Γ, such that for every Γ and partition $(Z_j : 1 \le j \le p)$ of Z the following is true. Suppose that*

(i) *if $F \subseteq \Sigma^*$ is a Γ-normal O-arc with $|F \cap V(\Gamma)| \le N$ then F bounds a closed disc $\Delta \subseteq \Sigma$ which includes at most one of $\Delta_1, \dots, \Delta_t$,*

(ii) *for every triple F_1, F_2, $F_3 \subseteq \Sigma^*$ such that F_1, F_2 are Γ-normal O-arcs bounding disjoint closed discs in Σ which each include exactly one of $\Delta_1, \dots, \Delta_t$, and F_3 is a Γ-normal line with an end in F_1 and an end in F_2 and no other point in $F_1 \cup F_2$, we have $|(F_1 \cup F_2 \cup F_3) \cap V(\Gamma)| > N$*

(iii) *for every pair F_1, $F_2 \subseteq \Sigma^*$ of Γ-normal O-arcs bounding closed discs D_1, $D_2 \subseteq \Sigma$ which each include exactly one of $\Delta_1, \dots, \Delta_t$, and with $|D_1 \cap D_2| = 1$, we have $|(F_1 \cup F_2) \cap V(\Gamma)| > N$*

(iv) *if Σ is a sphere and $t = 3$, and F_1, F_2, $F_3 \subseteq \Sigma^*$ are Γ-normal lines which have common ends but are otherwise mutually disjoint, and the O-arcs $F_2 \cup F_3$, $F_3 \cup F_1$, $F_1 \cup F_2$ respectively bound closed discs D_1, D_2, $D_3 \subseteq \Sigma$, say, where $F_i \not\subseteq D_i$ and $\Delta_i \subseteq D_i$ $(i = 1, 2, 3)$, then $|(F_1 \cup F_2 \cup F_3) \cap V(\Gamma)| > N$*

(v) *for every Γ-normal O-arc $F \subseteq \Sigma^*$ which bounds a closed disc in Σ including exactly one of $\Delta_1, \dots, \Delta_t$, ($\Delta_i$ say) we have $|F \cap V(\Gamma)| \ge |bd(\Delta_i) \cap V(\Gamma)|$, and*

(vi) *the topological feasibility condition is satisfied.*

Then (*) *holds.*

We wish to prove a more compact form of (3.1). If $\mathscr{T}$ is a tangle of order θ in Γ, we say that $X \subseteq V(\Gamma)$ is *free* (with respect to $\mathscr{T}$) if $|X| \le \theta$ and there [p. 246] is no $(A, B) \in \mathscr{T}$ of order $< |X|$ with $X \subseteq V(A)$. Our revised form of (3.1) is the following.

(3.2) *For every connected surface Σ and all integers $t \geq 0$ and $z \geq 0$, there exists an integer $\theta \geq 1$ such that the following is true. Let $\Delta_1, \ldots, \Delta_t$ be mutually disjoint closed discs in Σ, and let Γ be a 2-cell drawing in Σ such that $U(\Gamma) \cap \Delta_i = V(\Gamma) \cap bd(\Delta_i)$ for $1 \leq i \leq t$. Let $|Z| \leq z$, where*

$$Z = \bigcup_{1 \leq i \leq t} V(\Gamma) \cap \mathrm{bd}(\Delta_i),$$

and let $(Z_j : 1 \leq j \leq p)$ be a partition of Z satisfying the topological feasibility condition above. Let $\mathscr{T}$ be a respectful tangle of order $\geq \theta$ in Γ, with metric d, such that

(i) *$d(r_i, r_j) \geq \theta$ for $1 \leq i < j \leq t$, where r_i is the region of Γ meeting Δ_i for $1 \leq i \leq t$, and*

(ii) *$V(\Gamma) \cap bd(\Delta_i)$ is free for $1 \leq i \leq t$.*

Then (*) holds.

Proof. We assume first that either Σ is not a sphere or $t \geq 3$. Choose N satisfying (3.1) and let $\theta = 2N + 1$. We shall verify (3.1) (i)-(vi). Let K be a radial drawing of Γ. First, we observe:

(1) *Let $\phi : S^1 \to \Sigma$ be continuous, of length $< \theta$, with $|\phi|$ Γ-normal and $|\phi| \subseteq \Sigma^*$. Let F_1, $F_2 \subseteq |\phi|$ be O-arcs, and let $\Delta_i \subseteq ins(F_1)$, $\Delta_j \subseteq ins(F_2)$, say, where $1 \leq i, j \leq t$. Then $i = j$.*

Subproof. Let v_i, $v_j \in V(K)$ with $v_i \in r_i$, $v_j \in r_j$. Let W be a trace of ϕ. Now $\Delta_i \subseteq \mathrm{ins}(F_1)$ and so $r_i \cap \mathrm{ins}(F_1) \neq \emptyset$; therefore, $r_i \cap \mathrm{ins}(W) \neq \emptyset$ by (2.1). Similarly $r_j \cap \mathrm{ins}(W) \neq \emptyset$, and so $d(r_i, r_j) < \theta$. From hypothesis (i) we deduce that $i = j$. This proves (1).

(2) *If $F \subseteq \Sigma^*$ is a Γ-normal O-arc with $|F \cap V(\Gamma)| < \theta$ and F bounds a closed disc $\Delta \subseteq \Sigma$ which includes at most one of $\Delta_1, \ldots, \Delta_t$, then $\Delta = ins(F)$.*

Subproof. By (1), $\mathrm{ins}(F)$ is such a disc. If there is another, then Σ is a sphere and $t \leq 2$, a contradiction. This proves (2).

Now we verify (3.1)(i)-(vi). Let F be as in (3.1)(i). Let $\Delta = \mathrm{ins}(F)$; then by (1), Δ includes at most one of $\Delta_1, \ldots, \Delta_t$. This verifies (3.1)(i). For (3.1)(ii), (iii), suppose that one of them is false. Then there is a continuous $\phi : S^1 \to \Sigma$ of length $\leq 2N$, with $|\phi|$ Γ-normal and $|\phi| \subseteq \Sigma^*$, and there are F_1, F_2, Δ_i, Δ_j as in (1), where $i \neq j$. But this contradicts (1), since $\theta > 2N$.

To verify (3.1)(iv) let F_1, F_2, F_3, D_1, D_2, D_3 be as in (3.1)(iv). By (2), $D_1 = \mathrm{ins}(F_2 \cup F_3)$, $D_2 = \mathrm{ins}(F_3 \cup F_1)$, $D_3 = \mathrm{ins}(F_1 \cup F_2)$. But then $D_1 \cup D_2 \cup D_3 = \Sigma$ contrary to the second tangle axiom. To verify (3.1)(v), let F [p. 247] be as in (3.1)(v), bounding a closed disc $\Delta \subseteq \Sigma$, say, including

exactly one of $\Delta_1, \ldots, \Delta_t$, say Δ_i. Then $\Delta = \mathrm{ins}(F)$ by (2), and so

$$(\Gamma \cap \Delta_i, \Gamma \cap \overline{\Sigma - \Delta_i}) \in \mathscr{T};$$

but since $V(\Gamma) \cap \mathrm{bd}(\Delta_i)$ is free and $V(\Gamma) \cap \mathrm{bd}(\Delta_i) \subseteq V(\Gamma \cap \Delta_i)$, it follows that

$$|V(\Gamma) \cap \mathrm{bd}(\Delta_i)| \leq |V(\Gamma \cap \Delta_i \cap \overline{\Sigma - \Delta_i})| = |F \cap V(\Gamma)|,$$

as required. Finally, (3.1)(vi) holds by hypothesis. Thus (*) holds, by (3.1).

It remains to deal with the case when Σ is a sphere and $t \leq 2$. If $t = 0$ the result is vacuous. Let us assume $t = 2$. Choose θ' to satisfy (3.2) with Σ, z unchanged but with $t = 3$. Let $\theta = 2\theta' + 1$. By hypothesis (i), $d(r_1, r_2) \geq \theta = 2\theta' + 1$. Let $a_1, a_2, \ldots, a_s$ be a sequence of regions of Γ with $a_1 = r_1$, $a_s = r_2$, such that a_i and a_{i+1} are both incident with some vertex for $1 \leq i < s$. Then

$$|d(r_1, a_{i+1}) - d(r_1, a_i)| \leq d(a_i, a_{i+1}) \leq 2 \quad (1 \leq i < s)$$

and so we may choose i with $1 \leq i \leq s$ such that $0 \leq d(r_1, a_i) - \theta' \leq 1$. Now

$$d(r_2, a_i) \geq d(r_1, r_2) - d(r_1, a_i) \geq 2\theta' + 1 - (\theta' + 1) = \theta'.$$

But also $d(r_1, a_i) \geq \theta'$. Thus

$$d(r_1, a_i), d(r_2, a_i), d(r_1, r_2) \geq \theta'.$$

Let us take $r_3 = a_i$, and $\Delta_3 \subseteq r_3$ some closed disc, and apply the theorem proved above, that is, (3.2) with $t = 3$ and Σ a sphere. Certainly $V(\Gamma) \cap \mathrm{bd}(\Delta_3) = \emptyset$ and, hence, is free; we deduce that the desired $H_1, \ldots, H_p$ exist.

Finally, we deal with the case $t = 1$, when Σ is a sphere. Take $\theta = z$. By [2, Theorem (3.6)] it suffices to prove that for every Γ-normal line F_3 with ends $a, b \in \mathrm{bd}(\Delta_1)$ but with no other point in Δ_1, we have

$$|F_3 \cap V(\Gamma)| \geq \min(|F_1 \cap V(\Gamma)|, |F_2 \cap V(\Gamma)|),$$

where F_1, F_2 are the closures of the two components of $\mathrm{bd}(\Delta_1) - \{a, b\}$. Suppose then that F_3 is such a line.

Let $F_2 \cup F_3$, $F_1 \cup F_3$ bound closed discs D_1, $D_2 \subseteq \Sigma$ respectively, where $D_i \cap F_i = \{a, b\}$ $(i = 1, 2)$. By the second tangle axiom, it is not the case that

$$(\Gamma \cap D_i, \Gamma \cap \overline{\Sigma - D_i}) \in \mathscr{T} \quad (i = 1, 2),$$

since $(\Gamma \cap D_1) \cup (\Gamma \cap D_2) = \Gamma$. Thus we may assume that $(\Gamma \cap D_1, \Gamma \cap \overline{\Sigma - D_1}) \notin \mathscr{T}$. On the other hand, $Z \subseteq V(\Gamma \cap \overline{\Sigma - D_1})$ and Z is free, and

so if $(\Gamma \cap \overline{\Sigma - D_1}, \Gamma \cap D_1) \in \mathscr{T}$, then this separation has order $\geq |Z|$. If it does not [p. 248] belong to $\mathscr{T}$ then it has order $\geq \theta = |Z|$ by the first tangle axiom. Thus in either case it has order $\geq |Z|$, that is,

$$|(F_2 \cup F_3) \cap V(\Gamma)| \geq |(F_1 \cup F_2) \cap V(\Gamma)|.$$

Hence,

$$|F_3 \cap V(\Gamma)| \geq |F_1 \cap V(\Gamma)| \geq \min(|F_1 \cap V(\Gamma)|, |F_2 \cap V(\Gamma)|),$$

as required. □

4. Clique Minors

It is convenient to prove here three applications of (3.2) that we shall need in future papers. To do so we shall need the following.

(4.1) *Let $\mathscr{T}$ be a respectful tangle in a 2-cell drawing Γ in a connected surface Σ, with metric d. Let $u, v \in V(\Gamma)$, and let P be a path of Γ between u and v. If $(A, B) \in \mathscr{T}$ and $V(P) \subseteq V(A)$ then (A, B) has order $\geq \frac{1}{2}d(u, v)$.*

Proof. By [4, Theorem (2.9)], we may assume that $P \subseteq A$ and, clearly, that $E(\Gamma) \neq \emptyset$. By (1.1), there are circuits $C_1, \ldots, C_k$ of a radial drawing K of Γ, such that $V(C_i) \cap V(\Gamma) \subseteq V(A \cap B)$ $(1 \leq i \leq k)$ and such that every edge of A is a subset of one of $\mathrm{ins}(C_1), \ldots, \mathrm{ins}(C_k)$. Now since $U(P) \subseteq \mathrm{ins}(C_1) \cup \cdots \cup \mathrm{ins}(C_k)$, we may renumber $C_1, \ldots, C_k$ so that $u \in \mathrm{ins}(C_1)$, $v \in \mathrm{ins}(C_j)$ for some j, $\mathrm{ins}(C_i) \cap \mathrm{ins}(C_{i+1}) \neq \emptyset$ for all i with $1 \leq i < j$, and $\mathrm{ins}(C_i) \cap \mathrm{ins}(C_{i'}) = \emptyset$ for all i, i' with $1 \leq i, i' \leq j$ and $i' > i + 1$. Then

$$d(u, v) \leq \frac{1}{2}(|E(C_1)| + \cdots + |E(C_j)|) = \sum_{1 \leq i \leq j} |V(C_i) \cap V(\Gamma)|.$$

But no vertex of Γ belongs to more than two of $V(C_1), \ldots, V(C_j)$, and so

$$\sum_{1 \leq i \leq j} |V(C_i) \cap V(\Gamma)| \leq 2|V(C_1 \cup \cdots \cup C_j) \cap V(\Gamma)| \leq 2|V(A \cap B)|.$$

The result follows. □

Let G' be a subgraph of a graph G, and let $\mathscr{T}'$ be a tangle in G' of order θ. Let $\mathscr{T}$ be the set of all separations (A, B) of G of order $< \theta$ such that $(A \cap G', B \cap G') \in \mathscr{T}'$. Evidently $\mathscr{T}$ is a tangle in G of order θ, and we call it the tangle *induced by* $\mathscr{T}'$.

[p. 249] Let $\mathscr{T}$ be a tangle in a graph G, and let L be a graph. We say that *$\mathscr{T}$ controls an L minor of G* if there is a function α with domain $V(L) \cup E(L)$ such that

(i) for each $v \in V(L)$, $\alpha(v)$ is a non-null connected subgraph of G, and $\alpha(u), \alpha(v)$ are disjoint for all distinct $u, v \in V(L)$

(ii) for each $e \in E(L)$ with distinct ends u, v, $\alpha(e)$ is an edge of G with ends in $V(\alpha(u))$ and in $V(\alpha(v))$

(iii) for each loop $e \in E(L)$ with end v, $\alpha(e)$ is an edge of G not in $E(\alpha(v))$ with both ends in $V(\alpha(v))$

(iv) for distinct $e, f \in E(L)$, $\alpha(e) \neq \alpha(f)$

(v) $\mathscr{T}$ has order $\geq |V(L)|$, and there is no $(A, B) \in \mathscr{T}$ of order $< |V(L)|$ such that $V(\alpha(v)) \subseteq V(A)$ for some $v \in V(L)$.

(The quantity $|V(L)|$ in (v) is somewhat arbitrary, but will do for our applications. We remark that it could be replaced by any other increasing function of L, and (4.3)–(4.5) would remain true.)

We observe the following.

(4.2) *If L is isomorphic to a minor of L', and $\mathscr{T}$ is a tangle in a graph G, and $\mathscr{T}$ controls an L' minor of G then $\mathscr{T}$ controls an L minor of G.*

The proof is clear~~, and we omit it~~, and we give it here.

Let α be a function showing that $\mathscr{T}$ controls an L' minor of G. First, for each edge f of L', the edge $\alpha(f)$ belongs to no $\alpha(w)$ $(w \in V(L'))$. For suppose that $\alpha(f) \in E(\alpha(w))$. Then both ends of $\alpha(f)$ are in $V(\alpha(w))$. If f has distinct ends u, u', then by (ii) $\alpha(w)$ meets both $\alpha(u)$ and $\alpha(u')$, and so $w = u$ and $w = u'$ by (i), which is impossible. If f is a loop with end u, then by (iii) $\alpha(w)$ meets $\alpha(u)$, and so $w = u$ by (i), contrary to (iii).

Since L is isomorphic to a minor of L', there are mutually disjoint non-empty sets $Y(v) \subseteq V(L')$ $(v \in V(L))$, and distinct edges $\eta(e)$ of L' $(e \in E(L))$, with the following properties. If $e \in E(L)$ has distinct ends u, v, then $\eta(e)$ has one end in $Y(u)$ and the other in $Y(v)$. If $e \in E(L)$ is a loop with end v, then both ends of $\eta(e)$ are in $Y(v)$. For each $v \in V(L)$, the subgraph $M(v)$ of L' with vertex set $Y(v)$ and edge set all edges of L' with both ends in $Y(v)$ which are not of the form $\eta(e)$ is connected. (Take for $Y(v)$ the set of vertices of L' contracted into v, and for $\eta(e)$ the edge of L' that becomes e.)

For $v \in V(L)$, let $\alpha'(v)$ be the union of the subgraphs $\alpha(w)$ $(w \in Y(v))$, together with the edges $\alpha(f)$ $(f \in E(M(v)))$ and their ends. For $e \in E(L)$, let $\alpha'(e) = \alpha(\eta(e))$. We claim that α' shows that $\mathscr{T}$ controls an L minor of G. For (i), $\alpha'(v)$ is non-null, and it is connected, since each $\alpha(w)$ is connected, $M(v)$ is connected, and for each edge f of $M(v)$ the edge $\alpha(f)$ joins $\alpha(w)$ and $\alpha(w')$, where w, w' are the ends of f, by (ii) and (iii) for α. For distinct $u, v \in V(L)$, the subgraphs $\alpha'(u)$ and $\alpha'(v)$ are disjoint, since $Y(u) \cap Y(v) = \emptyset$, and since the subgraphs $\alpha(w)$ $(w \in V(L'))$ are mutually

disjoint and include all ends of the edges $\alpha(f)$. For (ii), let e have distinct ends u, v. Then $\eta(e)$ has ends $w \in Y(u)$ and $w' \in Y(v)$, and so by (ii) for α, $\alpha'(e)$ has ends in $V(\alpha(w)) \subseteq V(\alpha'(u))$ and in $V(\alpha(w')) \subseteq V(\alpha'(v))$. For (iii), let e be a loop with end v. Both ends of $\eta(e)$ are in $Y(v)$, and so by (ii) and (iii) for α, both ends of $\alpha'(e)$ are in $V(\alpha'(v))$. Moreover $\alpha'(e) \notin E(\alpha'(v))$, because $\alpha'(e)$ belongs to no $\alpha(w)$, as we saw, and $\alpha'(e) \neq \alpha(f)$ for $f \in E(M(v))$ by (iv) for α, since $f \neq \eta(e)$. For (iv), if $e, e' \in E(L)$ are distinct, then $\eta(e) \neq \eta(e')$, and so $\alpha'(e) \neq \alpha'(e')$ by (iv) for α. For (v), the sets $Y(v)$ ($v \in V(L)$) are mutually disjoint and non-empty, and so $|V(L)| \leq |V(L')|$. Hence $\mathscr{T}$ has order $\geq |V(L)|$, by (v) for α. Let $(A, B) \in \mathscr{T}$ have order $< |V(L)|$, and suppose that $V(\alpha'(v)) \subseteq V(A)$ for some $v \in V(L)$. Choose $w \in Y(v)$. Then $V(\alpha(w)) \subseteq V(\alpha'(v)) \subseteq V(A)$, and (A, B) has order $< |V(L')|$, contrary to (v) for α. This proves (4.2).[F5]

(4.3) *For any connected surface Σ and any graph L which can be drawn in Σ, there is an integer $\theta \geq 1$ such that the following is true. Let G be a graph, let Γ be a subgraph of G which is a 2-cell drawing in Σ, and let $\mathscr{T}$ be a respectful tangle in Γ of order $\geq \theta$. Then the tangle in G induced by $\mathscr{T}$ controls an L minor of G.*

Proof. Choose a graph L' which can be drawn in Σ, with a minor isomorphic to L, such that L' is loopless and every vertex of L' is incident with ≥ 2 edges of L'. By (4.2), if the result holds for L' then it holds for L. We may therefore assume that L is loopless and every vertex of L is incident with ≥ 2 edges. Let $t = |E(L)|$ and choose $\theta_1 \geq 2|V(L)|$ to satisfy (3.2) (with $z = 2t$). We claim that $\theta = 5 + t(\theta_1 + 7)$ satisfies (4.3). For let G, Γ, $\mathscr{T}$ be as in (4.3). It suffices to show that $\mathscr{T}$ controls an L minor of Γ. Let $E(L) = \{e_1, \ldots, e_t\}$.

An edge f of G with ends u, v is free if $u \neq v$ and $\{u, v\}$ is free. Let $g_1 \in E(\Gamma)$. By (1.2), there is an edge $g_2 \in E(\Gamma)$ such that $d(g_1, g_2) \geq \theta$. Let P be a path of Γ from v_1 to v_2, where v_i is an end of g_i ($i = 1, 2$).

(1) *For each $f \in E(P)$, if $d(g_1, f), d(g_2, f) \geq 2$ then f is free.*

Subproof. Suppose that f is not free. Let f have ends u, v. Certainly $u \neq v$ since P is a path, and so there exists $(A, B) \in \mathscr{T}$ of order ≤ 1 with [p. 250] $u, v \in V(A)$. By [4, Theorem (2.9)] we may assume that $f \in E(A)$. By (1.1) there is a circuit C of a radial graph K of Γ with $|E(C)| \leq 2$ and with $f \subseteq \mathrm{ins}(C)$. Let P_1, P_2 be the maximal subpaths of P not using f, with $v_1 \in V(P_1), v_2 \in V(P_2)$. Since $|U(C) \cap V(\Gamma)| \leq 1$ and $V(P_1) \cap V(P_2) = \emptyset$, we may assume that $U(C) \cap V(P_1) = \emptyset$. Since P_1 has one end in $\mathrm{ins}(C)$

[F5] **Missing proof.** The paper omits this proof. The argument above is the one used in the formalization.

it follows that $v_1 \in U(P_1) \subseteq \text{ins}(C) - U(C)$, and so $g_1 \subseteq \text{ins}(C)$. But $f \subseteq \text{ins}(C)$, and so $d(f, g_1) \leq 1$. This proves (1).

Now g_1, g_2 have no common end, since $d(g_1, g_2) \geq \theta \geq 5$. Thus $E(P) \neq \emptyset$; and if f is the edge of P incident with v_2, then $d(f, g_2) \leq 4$, and so $d(g_1, f) \geq \theta - 4$. Consequently for $1 \leq i \leq \theta - 4$ there is a first edge f_i of P (that is, nearest to v_1) with $d(g_1, f_i) \geq i$.

(2) *For* $1 \leq i \leq \theta - 4$, $i \leq d(g_1, f_i) \leq i + 3$.

Subproof. If f_i is incident with v_1, then $d(g_1, f_i) \leq 4 \leq i + 3$. If f_i is not incident with v_1, let f be the edge of P preceding f_i. From the definition of f_i, $d(g_1, f) \leq i - 1$, but $d(f, f_i) \leq 4$, and so $d(g_1, f_i) \leq i + 3$. This proves (2).

(3) *For* $1 \leq i < j \leq \theta - 4$, *if* $j - i \geq \theta_1 + 7$ *then* $d(f_i, f_j) \geq \theta_1 + 4$.

Subproof. $d(g_1, f_j) \geq j$ and $d(g_1, f_i) \leq i + 3$ by (2), and so $d(f_i, f_j) \geq j - i - 3 \geq \theta_1 + 4$.

(4) *For* $2 \leq i \leq \theta - 5$, f_i *is free.*

Subproof. $d(g_1, f_i) \geq i \geq 2$, and $d(g_2, f_i) \geq d(g_1, g_2) - d(g_1, f_i) \geq \theta - i - 3 \geq 2$, and so f_i is free by (1).

Let $h_i = f_{(\theta_1+7)i}$ $(i = 1, \ldots, t)$. Since $(\theta_1 + 7)t = \theta - 5$ it follows from (3) and (4) that $h_1, \ldots, h_t$ are all free, and $d(h_i, h_j) \geq \theta_1 + 4$ for $1 \leq i < j \leq t$. Let h_i have ends a_i, b_i $(1 \leq i \leq t)$. For $1 \leq i \leq t$ let $\Delta_i \subseteq \Sigma$ be a closed disc with $\Delta_i \cap U(\Gamma) = \{a_i, b_i\} \subseteq \text{bd}(\Delta_i)$. Since $d(h_i, h_j) \geq \theta_1 + 4$ for $i \neq j$, $\Delta_1, \ldots, \Delta_t$ are mutually disjoint. Let $Z = \{a_1, b_1, \ldots, a_t, b_t\}$. Direct the edges of L, arbitrarily, and for each $v \in V(L)$ let

$$Z_v = \{a_i; v \text{ is the head of } e_i\} \cup \{b_i; v \text{ is the tail of } e_i\}.$$

Then $\{Z_v : v \in V(L)\}$ is a partition of Z. We wish to apply (3.2) to $\Gamma, \mathscr{T}$.

(5) $d(r_i, r_j) \geq \theta_1$ *for* $1 \leq i < j \leq t$, *where* r_i *is the region of* Γ *meeting* Δ_i.

Subproof. $d(r_i, h_i) \leq 2$, since both ends of h_i are incident with r_i, and similarly $d(r_j, h_j) \leq 2$. Since $d(h_i, h_j) \geq \theta_1 + 4$, it follows that $d(r_i, r_j) \geq \theta_1$, as required.

(6) *For* $1 \leq i \leq t$, $V(\Gamma) \cap bd(\Delta_i)$ *is free with respect to* $\mathscr{T}$.

Subproof. $V(\Gamma) \cap \text{bd}(\Delta_i) = \{a_i, b_i\}$, and this is free since h_i is free.

[p. 251] (7) *The topological feasibility condition is satisfied.*

Subproof. Take a drawing of the directed graph L in Σ and replace each vertex v by a small closed disc D_v. By applying a homeomorphism of Σ which maps the residue of each edge of L onto h_i with the appropriate orientation, the images of the D_v's under this homeomorphism satisfy the topological feasibility condition.

From (5), (6), (7), and (3.2) we deduce that for each $v \in V(L)$ there is a connected subgraph $\alpha(v)$ of G with $Z \cap V(\alpha(v)) = Z_v$, such that $\alpha(u), \alpha(v)$ are mutually disjoint for all distinct u, v. For $1 \le i \le t$ define $\alpha(e_i) = h_i$. Then α satisfies conditions (i)–(iv) in the definition of "controls an L minor," and we claim that it also satisfies (v). For certainly $\theta \ge |V(L)|$; suppose that $(A, B) \in \mathscr{T}$, and $V(\alpha(v)) \subseteq V(A)$ for some $v \in V(L)$. By our assumption, there exist distinct i, j such that v is incident with e_i and e_j, and so Z_v meets both $\{a_i, b_i\}$ and $\{a_j, b_j\}$. Consequently there exist $c_i, c_j \in Z_v$ with $d(c_i, c_j) \ge \theta_1$, since $d(a_i, h_i), d(b_i, h_i) \le 2$ and $d(a_j, h_j), d(b_j, h_j) \le 2$, and $d(h_i, h_j) \ge \theta_1 + 4$. Since $c_i, c_j \in Z_v \subseteq V(\alpha(v))$ and $\alpha(v)$ is connected and $V(\alpha(v)) \subseteq V(A)$, we deduce that there is a path Q of Γ with ends c_i, c_j and with $V(Q) \subseteq V(A)$. By (4.1), (A, B) has order $\ge \frac{1}{2} d(c_i, c_j) \ge \frac{1}{2}\theta_1 \ge |V(L)|$. Hence (v) holds, and the proof is complete. □

(4.4) *For any connected surface Σ and any integer $p \ge 0$, there is an integer $\theta \ge 1$ such that the following is true. Let Γ be a 2-cell drawing in Σ, with a respectful tangle $\mathscr{T}$ of order $\ge \theta$. Let Γ be a subgraph of a graph G, let $q = \frac{1}{2}p(p-1)$, and for $1 \le i \le q$, let P_i be a path of G with distinct ends $u_i, v_i \in V(\Gamma)$ and with no other vertex or edge in Γ. Let $P_1, \ldots, P_q$ be mutually disjoint, and for all distinct $u, v \in \{u_1, v_1, \ldots, u_q, v_q\}$ let $d(u, v) \ge \theta$, where d is the metric of $\mathscr{T}$. Then the tangle in G induced by $\mathscr{T}$ controls a K_p minor of G.*

Proof. We may assume that $p \ge 5$, for otherwise K_p can be drawn in Σ and the result follows from (4.3). Let us choose $\theta' \ge 2p$ to satisfy (3.2) for Σ, taking $t = p(p-1)$ and $z = p(p-1)$. We claim that $\theta = \theta' + 2$ satisfies (4.4). For let Γ, $\mathscr{T}$, $P_1, \ldots, P_q$ be as in the theorem. We may renumber $P_1, \ldots, P_q$ as P_{ij} $(1 \le i < j \le p)$; and let P_{ij} have ends u_{ij}, v_{ij} $(1 \le i < j \le p)$. Let $Z = \{u_{ij}, v_{ij} : 1 \le i < j \le p\}$, and for $1 \le j \le p$ let

$$Z_j = \{u_{ij} : 1 \le i < j\} \cup \{v_{ij} : j < i \le p\}.$$

Then $|Z| = p(p-1)$, and $(Z_1, \ldots, Z_p)$ is a partition of Z. For each $v \in Z$, let $\Delta_v \subseteq \Sigma$ be a closed disc with $v \in \mathrm{bd}(\Delta_v)$ and $U(\Gamma) \cap \Delta_v = \{v\}$, chosen all mutually disjoint. Let $\Delta_v \subseteq \{v\} \cup r_v$, where r_v is a region. Since $V(\Gamma) \cap \mathrm{bd}(\Delta_v)$ is free for each v (because Γ is connected) and the topological feasibility condition is trivially true (because $|V(\Gamma) \cap \mathrm{bd}(\Delta_v)| = 1$ for each v), and $d(r_u, r_v) \ge d(u, v) - 2 \ge \theta - 2 = \theta'$, it follows from (3.2) that there [p. 252] are disjoint connected subgraphs $H_1, \ldots, H_p$ of Γ with $V(H_j) \cap Z = Z_j$ $(1 \le j \le p)$.

For $1 \le i < j \le p$, let Q_{ij} be the path obtained from P_{ij} by deleting v_{ij}, and let e_{ij} be the edge of P_{ij} incident with v_{ij}. For $1 \le j \le p$, let

$X_j = H_j \cup \bigcup(Q_{ij} : 1 \le i < j)$. We see that $X_1, \ldots, X_p$ are mutually disjoint connected subgraphs of G, and for $1 \le i < j \le p$ there is an edge of G between $V(X_i)$ and $V(X_j)$, namely, e_{ij}.

Let $\mathscr{T}^*$ be the tangle in G induced by $\mathscr{T}$, let $(A, B) \in \mathscr{T}^*$, and suppose that $V(X_j) \subseteq V(A)$ for some j. Since $p \ge 5 \ge 3$, it follows that $|Z_j| \ge 2$; choose $u, v \in Z_j$, distinct. Then $d(u, v) \ge \theta$. Now $u, v \in Z_j \subseteq V(H_j)$, and H_j is connected; let P be a path of H_j with ends u, v. Then $V(P) \subseteq V(H_j) \subseteq V(A \cap \Gamma)$, and $(A \cap \Gamma, B \cap \Gamma) \in \mathscr{T}$ by definition of $\mathscr{T}^*$, and so by (4.1), $(A \cap \Gamma, B \cap \Gamma)$ has order $\ge \frac{1}{2}d(u, v) \ge \frac{1}{2}\theta \ge p$, and hence so does (A, B). It follows that $\mathscr{T}^*$ controls a K_p minor. $\square$

(4.5) *Let Σ be a connected surface, and let $p \ge 0$ be an integer. Then there are integers $\theta \ge 1$ and $t \ge 0$ such that the following is true. Let Γ be a 2-cell drawing in Σ, with a respectful tangle $\mathscr{T}$ of order $\ge \theta$, with metric d. Let $r_1, \ldots, r_t$ be distinct regions of Γ such that $d(r_i, r_j) \ge \theta$ for $1 \le i < j \le t$. For $1 \le i \le t$, let C_i be a circuit of Γ such that $U(C_i)$ bounds r_i; and let $a_i, b_i, c_i, d_i \in V(C_i)$ be distinct, so that a_i, b_i, c_i, d_i occur in $V(C_i)$ in that order, and $\{a_i, b_i, c_i, d_i\}$ is free with respect to $\mathscr{T}$. Let Γ be a subgraph of a graph G, and for $1 \le i \le t$ let P_i, Q_i be paths of G with ends a_i, c_i and b_i, d_i, respectively, and with no other vertices or edges in Γ. Let $P_1, Q_1, \ldots, P_t, Q_t$ be mutually disjoint. Then the tangle in G induced by $\mathscr{T}$ controls a K_p minor.*

Proof. As in (4.4), we may assume that $p \ge 5$. Now K_p can be drawn with crossings in the plane, and hence in Σ, so that (i) no edge passes through any vertex, (ii) every two edges meet only finitely many times, (iii) whenever two edges meet they cross (that is, edges never touch "tangentially"), (iv) no three edges pass through any point of the surface, (v) no edge crosses itself, (vi) every edge crosses at least one other edge, and (vii) every vertex is incident with two edges that do not cross each other. To see this, first draw a Hamilton circuit without crossings, and then add the remaining edges to the drawing, so that (i)–(vi) are satisfied. Then (vii) is satisfied too.

Let this drawing have t crossings. Let L be obtained from K_p by subdividing its edges, where if $e \in E(K_p)$ is involved in k crossings of the drawing of K_p then e is subdivided $k - 1$ times. It follows that L can be drawn in Σ so that every edge is involved in exactly one crossing, and (i)–(vii) above are satisfied. Thus $t = \frac{1}{2}|E(L)|$, and the edges of L fall into t crossing pairs. Let the crossings be $x_1, \ldots, x_t$. By choosing a small disc around each crossing and applying a suitable homeomorphism, we deduce

[p. 253] (1) *Let $\Delta_1, \ldots, \Delta_t \subseteq \Sigma$ be disjoint closed discs, let $N_i \subseteq bd(\Delta_i)$ with $|N_i| = 4$ for $1 \le i \le t$, and let $Z = N_1 \cup \cdots \cup N_t$. Then there is a*

partition $(Z_v : v \in V(L))$ *of* Z *satisfying the topological feasibility condition of Section 3, such that for* $1 \le i \le p$*, if* $e, e' \in E(L)$ *cross at* x_i *and have ends* u, v *and* u', v'*, respectively in* L*, then the four members of* N_i *can be enumerated in order on* $bd(\Delta_i)$ *as* $\{a, b, c, d\}$*, where* $a \in Z_u, b \in Z_{u'}, c \in Z_v, d \in Z_{v'}$*.*

Let us choose θ to satisfy (3.2) for Σ and t, taking $z = 4t$. We claim that θ satisfies (4.5). For let $G, \Gamma, \mathscr{T}, r_i, C_i, a_i, b_i, c_i, d_i, P_i, Q_i$ $(1 \le i \le t)$ be as in (4.5). For $1 \le i \le t$ let $N_i = \{a_i, b_i, c_i, d_i\}$, and let $Z = N_1 \cup \cdots \cup N_t$. Let $(Z_v : v \in V(L))$ be as in (1). By (3.2), there are disjoint connected subgraphs H_v $(v \in V(L))$ of Γ such that $V(H_v) \cap Z = Z_v$ $(v \in V(L))$.

Let us renumber $P_1, Q_1, \ldots, P_t, Q_t$ as R_e $(e \in E(L))$, in such a way that if e is involved in x_i and has ends u, v, then R_e is one of P_i, Q_i and has ends in $V(H_u), V(H_v)$, respectively. (This is possible since $(Z_v : v \in V(L))$ satisfies (1).) For each $e \in E(L)$ let $\alpha(e) \in E(R_e)$. For each $v \in V(L)$ let $\alpha(v)$ be the component of

$$(\bigcup (H_v : v \in V(L)) \cup \bigcup (R_e : e \in E(L))) \setminus \{\alpha(e) : e \in E(L)\}$$

which includes H_v. Then it follows that if $e \in E(L)$ has ends u, v then $\alpha(e)$ has ends in $V(\alpha(u)), V(\alpha(v))$.

Consequently, conditions (i)–(iv) in the definition of "controls an L minor" are satisfied. For (v), let $\mathscr{T}^*$ be the tangle in G induced by $\mathscr{T}$. Then $\mathscr{T}^*$ has order $\ge \theta \ge 2t \ge |V(L)|$; let $(A, B) \in \mathscr{T}^*$ and suppose that $V(\alpha(v)) \subseteq V(A)$ for some $v \in V(L)$. There are two edges of L incident with v that do not cross each other, and so there exist i, j with $1 \le i < j \le t$ such that $Z_v \cap N_i, Z_v \cap N_j \ne \emptyset$. Choose $a_i \in Z_v \cap N_i, a_j \in Z_v \cap N_j$. Since $d(r_i, r_j) \ge \theta$ it follows that $d(a_i, a_j) \ge \theta - 2$, and so (applying (4.1) as in the proof of (4.3) and (4.4)) $(A \cap \Gamma, B \cap \Gamma) \in \mathscr{T}$ has order $\ge \frac{1}{2}(\theta - 2) \ge |V(L)|$. Consequently, $\mathscr{T}^*$ controls an L minor of G, and since L has a K_p minor, it follows from (4.2) that $\mathscr{T}^*$ controls a K_p minor of G, as required. □

5. Preserving Respect

The remainder of the paper is concerned with the effect on a respectful tangle and the associated metric of making local changes in the drawing or surface. The first local change we need to study is the following. Suppose we have a drawing in a surface, with a respectful tangle. Now we add a "handle" to the surface, attaching it on at two regions of the drawing, and we extend the drawing onto the handle. We know that the original tangle induces a tangle in the enlarged drawing; our question here is, under what

[p. 254] circumstances is the induced tangle respectful? We are also concerned with the parallel question when we add a crosscap.

Let Σ be a connected surface, and let $C_1, \dots, C_t$ be mutually disjoint O-arcs in Σ. Let Σ_1, Σ_2 be 2-manifolds with boundary, such that $\Sigma_1 \cup \Sigma_2 = \Sigma$ and

$$\Sigma_1 \cap \Sigma_2 = bd(\Sigma_1) = bd(\Sigma_2) = C_1 \cup \cdots \cup C_t.$$

Let $\hat{\Sigma}_1$ be a surface with $\Sigma_1 \subseteq \hat{\Sigma}_1$, such that each C_i bounds an open disc $D_i \subseteq \hat{\Sigma}_1$, and $D_1, \dots, D_t$ are mutually disjoint, and $\hat{\Sigma}_1 - \Sigma_1 = D_1 \cup \cdots \cup D_t$. If $t = 2$, $\hat{\Sigma}_1$ is connected, and Σ_2 is homeomorphic to $\{(x, y) : 1 \le x^2 + y^2 \le 2\}$ we say Σ_2 is a handle of Σ. Similarly, if $t = 1$, $\hat{\Sigma}_1$ is connected, and Σ_2 is homeomorphic to a Möbius band, we say that Σ_2 is a crosscap of Σ. We wish to prove the following two results.

(5.1) *Let Σ_2 be a handle of Σ, and let $\Sigma_1, \hat{\Sigma}_1$ be defined as above. Let the two components of $\Sigma_1 \cap \Sigma_2$ be C_1, C_2. Let Γ be a drawing in Σ such that C_1, C_2 are Γ-normal, and*

$$|V(\Gamma) \cap C_1| = |V(\Gamma) \cap C_2| = \theta,$$

and such that $\Gamma \cap \Sigma_2$ has θ components, each a path with one end in C_1 and the other in C_2. Suppose that every Γ-normal line $F \subseteq \Sigma_1$ with one end in C_1 and one end in C_2 satisfies $|F \cap V(\Gamma)| \ge \theta$. Let $\mathscr{T}_1$ be a tangle in $\Gamma \cap \hat{\Sigma}_1$, respectful towards $\hat{\Sigma}_1$, of order θ, such that $V(\Gamma) \cap C_1, V(\Gamma) \cap C_2$ are free with respect to $\mathscr{T}_1$. Then the tangle in Γ induced by $\mathscr{T}_1$ is respectful towards Σ.

(5.2) *Let Σ_2 be a crosscap of Σ, and let $\Sigma_1, \hat{\Sigma}_1$ be defined as above. Let Γ be a drawing in Σ such that $\Sigma_1 \cap \Sigma_2$ is Γ-normal, and $|V(\Gamma) \cap \Sigma_1 \cap \Sigma_2| = 2\theta$, and $\Gamma \cap \Sigma_2$ has θ components, each a path with ends some opposite pair of vertices in $V(\Gamma) \cap \Sigma_1 \cap \Sigma_2$, in the natural sense. Suppose that $\mathscr{T}_1$ is a tangle in $\Gamma \cap \hat{\Sigma}_1$, respectful towards $\hat{\Sigma}_1$, of order $\theta \ge 2$,[F6] and every subset $V(\Gamma) \cap \Sigma_1 \cap \Sigma_2$ of cardinality $\le \theta$ is free with respect to $\mathscr{T}_1$. Then the tangle in Γ induced by $\mathscr{T}_1$ is respectful towards Σ.*

Let us formulate a common generalization of (5.1) and (5.2).

(5.3) *Let Σ be a connected surface, and let $C_1, \dots, C_t$ be mutually disjoint O-arcs in Σ. Let Σ_1, Σ_2 be 2-manifolds with boundary, such that $\Sigma_1 \cup \Sigma_2 = \Sigma$*

[F6] **Degenerate case.** As printed, (5.2) fails for $\theta = 1$. Let Σ be the projective plane, cut into a closed disc Σ_1 and a Möbius band Σ_2. Let Γ have two opposite vertices u, v on $\Sigma_1 \cap \Sigma_2$, a path in Σ_1 joining them, and a path in Σ_2 joining them that misses the core circle of the Möbius band. The tangle of order 1 is respectful towards the sphere $\hat{\Sigma}_1$. But the core circle misses Γ and bounds no disc, so the induced tangle is not respectful towards Σ. For $\theta \ge 2$, (5.2) holds, and its use in Graph Minors XV has $\theta \ge 2$.

and

$$\Sigma_1 \cap \Sigma_2 = bd(\Sigma_1) = bd(\Sigma_2) = C_1 \cup \cdots \cup C_t.$$

Let $\hat{\Sigma}_1$ be a surface with $\Sigma_1 \subseteq \hat{\Sigma}_1$, such that each C_i bounds an open disc $D_i \subseteq \hat{\Sigma}_1$, and $D_1, \ldots, D_t$ are mutually disjoint, and $\hat{\Sigma}_1 - \Sigma_1 = D_1 \cup \cdots \cup D_t$.

[p. 255] *Let Γ be a drawing in Σ, such that $C_1 \cup \cdots \cup C_t$ is Γ-normal, and let $\Gamma_1 = \Gamma \cap \hat{\Sigma}_1$. Let $\mathscr{T}_1$ be a tangle in Γ_1 of order θ, respectful towards $\hat{\Sigma}_1$, and let $\mathscr{T}$ be a tangle in Γ induced by $\mathscr{T}_1$. Suppose that*

(i) *for $1 \le i < j \le t$, every Γ_1-normal line $F \subseteq \Sigma_1$ with one end in C_i and the other in C_j satisfies $|F \cap V(\Gamma_1)| \ge \theta$*

(ii) *for $1 \le i \le t$, $|C_i \cap V(\Gamma_1)| \ge \theta$ and every $X \subseteq C_i \cap V(\Gamma)$ with $|X| \le \theta$ is free with respect to $\mathscr{T}_1$, and*

(iii) *every continuous $\phi\colon S^1 \to \Sigma_2$ of length $< \theta$ with $|\phi|$ Γ-normal is null-homotopic in Σ_2.*

Then $\mathscr{T}$ is respectful towards Σ.

The proof of (5.3) uses the following lemma.

(5.4) *Under the same hypotheses, let F be a Γ-normal line with $F \subseteq \Sigma_1$, with ends in $C_1 \cup \cdots \cup C_t$, and with no other point in $C_1 \cup \cdots \cup C_t$, such that $|F \cap V(\Gamma)| < \theta$. Then there is a Γ-normal line $\tilde{F} \subseteq C_1 \cup \cdots \cup C_t$, with the same ends as F, such that $|\tilde{F} \cap V(\Gamma)| \le |F \cap V(\Gamma)|$, and such that there is a closed disc $\Delta \subseteq \Sigma_1$ bounded by $F \cup \tilde{F}$, with $(\Gamma_1 \cap \Delta, \Gamma_1 \cap \overline{\Sigma_1 - \Delta}) \in \mathscr{T}_1$.*

Proof. Let the ends of F be a, b, where $a \in C_i, b \in C_j$. Now $i = j$, from (5.3)(i). We assume $i = j = 1$ without loss of generality. Let P be a line with $P \subseteq D_1 \cup \{a, b\}$, with ends a, b. Then $P \cup F$ is a Γ_1-normal O-arc in $\hat{\Sigma}_1$, of length $< \theta$, and so there is a closed disc $\Delta_1 \subseteq \hat{\Sigma}_1$ bounded by $P \cup F$, satisfying:

(1) $(\Gamma_1 \cap \Delta_1, \Gamma_1 \cap \overline{\hat{\Sigma}_1 - \Delta_1}) \in \mathscr{T}_1$.

Let $\Delta_1 \cap C_1 = \tilde{F}$; then $\tilde{F}$ is a line with ends a, b and with $\tilde{F} \subseteq C_1 \cup \cdots \cup C_t$. Let Δ be the closed disc in Δ_1 bounded by the O-arc $F \cup \tilde{F}$. Then

$$(\Gamma_1 \cap \Delta, \Gamma_1 \cap \overline{\Sigma_1 - \Delta}) = (\Gamma_1 \cap \Delta_1, \Gamma_1 \cap \overline{\hat{\Sigma}_1 - \Delta_1}) \in \mathscr{T}_1$$

by (1). We claim that $\Delta \subseteq \Sigma_1$. For suppose not; then $D_i \cap \Delta \ne \emptyset$ for some $i \ge 2$, say $i = 2$, and so $D_2 \subseteq \Delta_1$, because $bd(\Delta_1) \subseteq \Sigma_1$. But $V(\Gamma_1) \cap C_2$ includes a subset of cardinality θ, which is free with respect to $\mathscr{T}_1$, by (ii); and yet $V(\Gamma_1) \cap C_2 \subseteq V(\Gamma_1 \cap \Delta_1)$, and the separation of (1) has order $|F \cap V(\Gamma_1)| < \theta$, a contradiction. Thus $\Delta \subseteq \Sigma_1$.

Finally, we claim that $|\tilde{F} \cap V(\Gamma_1)| \le |F \cap V(\Gamma_1)|$. For suppose not. Since $|F \cap V(\Gamma_1)| < \theta$, we may choose $X \subseteq \tilde{F} \cap V(\Gamma_1)$ with $|F \cap V(\Gamma_1)| < |X| \le \theta$. By (ii), X is free with respect to $\mathscr{T}_1$. But $X \subseteq V(\Gamma_1 \cap \Delta_1)$, and the separation

of (1) has order $|F \cap V(\Gamma_1)| < |X|$, a contradiction. Thus $\tilde{F}$ satisfies the lemma. □

[p. 256] *Proof of* (5.3). Let F be a Γ-normal O-arc in Σ with $|F \cap V(\Gamma)| < \theta$. We must show that F bounds a closed disc $\Delta \subseteq \Sigma$ with

$$(\Gamma \cap \Delta, \Gamma \cap \overline{\Sigma - \Delta}) \in \mathscr{T}.$$

We assume first that $|F \cap \Sigma_2| \leq 1$. Then $F \subseteq \hat{\Sigma}_1$, and so bounds a disc $\Delta \subseteq \hat{\Sigma}_1$, with

$$(\Gamma_1 \cap \Delta, \Gamma_1 \cap \overline{\hat{\Sigma}_1 - \Delta}) \in \mathscr{T}_1.$$

We claim that $\Delta \subseteq \Sigma$. For suppose that $\Delta \cap D_1 \neq \emptyset$, say. Then $D_1 \subseteq \Delta$, which is impossible since $V(\Gamma) \cap C_1$ includes a free subset of cardinality θ. Thus $\Delta \subseteq \Sigma$, and

$$(\Gamma \cap \Delta, \Gamma \cap \overline{\Sigma - \Delta}) \in \mathscr{T}$$

from the definition of $\mathscr{T}$.

Thus we may assume that $|F \cap \Sigma_2| \geq 2$. Let $f\colon S^1 \to F$ be a homeomorphism. Let the components of $F - \Sigma_2$ be X_i $(i \in I)$; then each X_i is homeomorphic to the open interval $(0,1)$. Let $\bar{X}_i = X_i \cup \{s_i, t_i\}$, where $s_i, t_i \in C_1 \cup \cdots \cup C_t$. By (5.4), there is a homeomorphism $f_i\colon \bar{X}_i \to C_1 \cup \cdots \cup C_t$ such that $f_i(s_i) = s_i$, $f_i(t_i) = t_i$, and

$$|\{x \in \bar{X}_i : f_i(x) \in V(\Gamma)\}| \leq |\bar{X}_i \cap V(\Gamma)|,$$

and such that there is a closed disc $\Delta_i \subseteq \Sigma_1$, bounded by $\bar{X}_i \cup \{f_i(x) : x \in \bar{X}_i\}$, such that

$$(\Gamma_1 \cap \Delta_i, \Gamma_1 \cap \overline{\Sigma_1 - \Delta_i}) \in \mathscr{T}_1.$$

For $x \in S^1$, define

$$g(x) = \begin{cases} f(x) & \text{if } f(x) \in \Sigma_2 \\ f_i(f(x)) & \text{if } f(x) \in X_i \quad (i \in I). \end{cases}$$

Then $g\colon S^1 \to \Sigma_2$ is continuous, and $|g|$ is Γ-normal, and g has length $\leq |F \cap V(\Gamma)| < \theta$. By (iii), g is null-homotopic in Σ_2, and so F is null-homotopic in $\Sigma_2 \cup \bigcup(\Delta_i : i \in I)$. By [1, Theorem (1.7)],[F7] there is a closed disc

$$\Delta \subseteq \Sigma_2 \cup \bigcup(\Delta_i : i \in I)$$

[F7] **Proof gap.** A gap in the proof: the set I may be infinite. If F touches $C_1 \cup \cdots \cup C_t$ from inside Σ_1 at a sequence of intervals that converge to a point p, without entering Σ_2, then $\Sigma_2 \cup \bigcup(\Delta_i : i \in I)$ need not be a 2-manifold at p, and [1, Theorem (1.7)] does not apply. The proof can be completed by first pushing such touching parts of F off Σ_2, inside a collar of $C_1 \cup \cdots \cup C_t$, so that only finitely many pieces remain. Then (5.3) holds as printed.

bounded by F. Since each $bd(\Delta_i)$ is Γ-normal,

$$\Gamma_1 \cap \Delta = \bigcup_{i \in I} (\Gamma_1 \cap \Delta_i) \cup (\Gamma_1 \cap F \cap \Sigma_1 \cap \Sigma_2).$$

[p. 257] Since $V(\Gamma_1)$, $E(\Gamma_1)$ are finite, there is a finite subset $I_0 \subseteq I$ such that

$$\Gamma_1 \cap \Delta = \bigcup_{i \in I_0} (\Gamma_1 \cap \Delta_i) \cup (\Gamma_1 \cap F \cap \Sigma_1 \cap \Sigma_2).$$

Let $I_0 = \{1, \ldots, k\}$, say. Let $(A, B) = (\Gamma_1 \cap \Delta, \Gamma_1 \cap \overline{\Sigma - \Delta})$. We wish to show that $(A, B) \in \mathscr{T}_1$. For let

$$\begin{aligned} (A_0, B_0) &= (\Gamma_1 \cap F \cap \Sigma_1 \cap \Sigma_2, \Gamma_1) \\ (A_i, B_i) &= (\Gamma_1 \cap \Delta_i, \Gamma_1 \cap \overline{\Sigma_1 - \Delta_i}) \quad (1 \leq i \leq k). \end{aligned}$$

Then for $0 \leq i \leq k$, $(A_i, B_i) \in \mathscr{T}$ and

$$V(A_i \cap B_i) \subseteq F \cap V(\Gamma_1) \subseteq V(A \cap B).$$

Also, $A = A_0 \cup A_1 \cup \cdots \cup A_k$. By [5, Theorem (6.2)], $(B, A) \notin \mathscr{T}_1$. Since (A, B) has order $|F \cap V(\Gamma)| < \theta$ it follows that $(A, B) \in \mathscr{T}_1$. We deduce, from the definition of $\mathscr{T}$, that

$$(\Gamma \cap \Delta, \Gamma \cap \overline{\Sigma - \Delta}) \in \mathscr{T},$$

as required. □

We see that (5.2) follows ~~immediately~~ from (5.3),[F8] and (5.3) also implies (5.1) because by [4, Theorem (12.2)], every subset of a free set is free.

6. Filling In A Disc

[F8] **Proof gap.** The deduction is not immediate. To apply (5.3) one needs its condition (iii), that every continuous $\phi\colon S^1 \to \Sigma_2$ of length $< \theta$ with $|\phi|$ Γ-normal is null-homotopic in Σ_2, and this does not follow at once from the hypotheses of (5.2). For $\theta = 1$ it is false: in the example in the note to (5.2), the core circle of the Möbius band Σ_2 misses Γ, so it has length 0, and it is not null-homotopic in Σ_2. For $\theta \geq 2$, (iii) holds. Let P be one of the θ paths of $\Gamma \cap \Sigma_2$. The ends of the paths are opposite pairs, so each other path has one end on each of the two arcs of $\Sigma_1 \cap \Sigma_2$ between the ends of P. Such a path joins the two arcs and misses P, so cutting Σ_2 along P leaves it connected, and a Möbius band cut along a path that leaves it connected is a closed disc. So $\Sigma_2 - U(P)$ is simply-connected. Now if $|\phi|$ met all θ paths, then, since the paths are disjoint, ϕ would have length at least θ. So $|\phi|$ misses one of the paths, say P, and ϕ is null-homotopic in $\Sigma_2 - U(P)$, and so in Σ_2. Hence (5.3) gives (5.2) for $\theta \geq 2$.

The problem of this section is somewhat similar to that of the last. Let Γ be a 2-cell drawing in a connected surface Σ, and let F be a Γ-normal O-arc bounding a closed disc $\Delta \subseteq \Sigma$. Let $|F \cap V(\Gamma)| = \theta \geq 2$, where θ is even, and let

$$F \cap V(\Gamma) = \{v_1, v_2, \ldots, v_\theta\}$$

numbered in order on F. Let the components of $\Gamma \cap \Delta$ be $P_1, P_2, \ldots, P_{\theta/2}$, where for $1 \leq i \leq \theta/2$ P_i is a path with ends $v_i, v_{\theta+1-i}$. Let e_i be an edge of P_i $(1 \leq i \leq \theta/2)$. Let $\Sigma_1 = \overline{\Sigma - \Delta}$, let $\Gamma_1 = \Gamma \cap \Sigma_1$ be 2-cell (in Σ), and let $\mathscr{T}_1$ be a tangle in Γ_1 of order $\theta_1 > \theta$, respectful towards Σ. Let $\mathscr{T}$ be the tangle in Γ induced by $\mathscr{T}_1$; then $\mathscr{T}$ is clearly respectful towards Σ. Let d be the metric derived from $\mathscr{T}$. Finally, let $F \cap V(\Gamma_1)$ be free with respect to $\mathscr{T}_1$.

[p. 258] (6.1) *In these circumstances, if $\theta \geq 4$ then $d(e_1, e_{\theta/2}) \geq \theta$.*[F9]

Proof. The lines $U(P_1), \ldots, U(P_{\theta/2})$ divide $\Delta - F$ into $\theta/2 + 1$ open discs $D_0, D_1, \ldots, D_{\theta/2}$, say, where $U(P_i) = \bar{D}_{i-1} \cap \bar{D}_i$ $(1 \leq i \leq \theta/2)$. For $0 \leq i \leq \theta/2$ let r_i be the region of Γ including D_i. Since Γ_1 is 2-cell it follows that $r_0, r_1, \ldots, r_{\theta/2}$ are all distinct. Hence we may choose a radial drawing K of Γ such that for $0 \leq i \leq \theta/2$, D_i contains a vertex v_i say of K, and D_i includes edges of K from v_i to every vertex of P_i (if $i \geq 1$) and to every vertex of P_{i+1} (if $i < \theta/2$). The metric d was defined on page 242 from some radial drawing of Γ, but it does not depend on which one. For if d' is the metric computed from another radial drawing, then (7.7)(ii), applied with $\Gamma' = \Gamma$ and $\mathscr{T}' = \mathscr{T}$ in both orders, gives $d \geq d'$ and $d' \geq d$; and the proof of (7.7) uses only (1.2) and (2.1), not (6.1). So we may compute d in K.[F10]

(1) $\mathrm{ins}(C) \subseteq \Delta$ *for every circuit C of K with $U(C) \subseteq \Delta$.*

Subproof. Let $D \subseteq \Delta$ be the closed disc bounded by $U(C)$ for some circuit C of K with $U(C) \subseteq \Delta$. Then there is no circuit of $\Gamma \cap D$, and so by [4, Theorem (2.10)], $(\Gamma \cap \overline{\Sigma - D}, \Gamma \cap D) \notin \mathscr{T}$. Hence $(\Gamma \cap D, \Gamma \cap \overline{\Sigma - D}) \in \mathscr{T}$, and so $D = \mathrm{ins}(C)$, as required.

Let W be a trace of a homeomorphism of S^1 onto F. Then $U(W) \subseteq \Delta$ from our choice of K, and W has length 2θ since $|F \cap V(\Gamma)| = \theta$. Since $\theta_1 > \theta$ it follows that $\mathrm{ins}(W)$ is defined, and by (1) $\mathrm{ins}(W) \subseteq \Delta$.

(2) *There is no closed walk W' of K of length $< 2\theta$ with* $\mathrm{ins}(W) \subseteq \mathrm{ins}(W')$, *such that $U(W')$ is included in the closure of* $\Sigma - \mathrm{ins}(W')$.

[F9] **Degenerate case.** As printed, (6.1) fails for $\theta = 2$: then $\theta/2 = 1$, so $e_1 = e_{\theta/2}$ and $d(e_1, e_1) = 0 < 2$. For even $\theta \geq 4$, (6.1) holds.

[F10] **Proof gap.** The paper computes d in this special K without saying why d is the same for every radial drawing. Graph Minors XI, page 75, says that a radial drawing is unique up to homeomorphisms of Σ fixing $U(\Gamma)$, but neither paper says how the metric changes under them. The inserted text gives the reason.

Subproof. Suppose that such W' exists. Let $\Delta' = \mathrm{ins}(W')$, $\Sigma' = \overline{\Sigma - \Delta'}$, so that $\Delta' \cap \Sigma' = U(W')$, $\Delta' \cup \Sigma' = \Sigma$. Let $A = \Gamma \cap \Delta'$, $B = \Gamma \cap \Sigma'$; then (A, B) is a separation of Γ of order

$$|V(\Gamma) \cap \Delta' \cap \Sigma'| = |V(\Gamma) \cap U(W')| < \theta$$

since W' has length $< 2\theta$. Now for every $e \in E(A)$, since $e \subseteq \mathrm{ins}(W')$ and $e \cap U(W') = \emptyset$, there is a circuit C of $K|W'$ with $e \subseteq \mathrm{ins}(C)$, and for such a circuit C,

$$V(C) \cap V(\Gamma) \subseteq U(W') \cap V(\Gamma) = \Delta' \cap \Sigma' \cap V(\Gamma) = A \cap B.$$

By (1.1), $(A, B) \in \mathscr{T}$, and so $(A \cap \Gamma_1, B \cap \Gamma_1) \in \mathscr{T}_1$. But

$$F \cap V(\Gamma_1) = U(W) \cap V(\Gamma) \subseteq V(A \cap \Gamma_1)$$

and $|F \cap V(\Gamma_1)| = \theta$, and $F \cap V(\Gamma_1)$ is free with respect to $\mathscr{T}_1$, a contradiction since $(A \cap \Gamma_1, B \cap \Gamma_1)$ has order $< \theta$. This proves (2).

Now there is no closed walk W^* of K of length $< 2\theta$ with $U(W^*) \subseteq \Delta$ and with $e_1, e_{\theta/2} \subseteq \mathrm{ins}(W^*)$. For every edge of K in Δ lies in some $\bar{D}_i$ and joins v_i to a vertex of P_i or P_{i+1} (since v_i is the vertex of K in r_i, and $r_i \cap \Delta \subseteq \bar{D}_i$). Give v_i the level $2i$ $(0 \le i \le \theta/2)$, and each vertex of P_i the level $2i-1$ $(1 \le i \le \theta/2)$; then each edge of K in Δ changes the level by one. Now let W^* be such a walk. Since $e_1 \cap U(W^*) = \emptyset$, there is a circuit C of $K|W^*$ with $e_1 \subseteq \mathrm{ins}(C)$, and $\mathrm{ins}(C) \subseteq \Delta$ by (1). Suppose that $v_0 \notin V(C)$. Then $U(C)$ is disjoint from D_0 and from the open arc of F in $\bar{D}_0$. But $\mathrm{ins}(C)$ meets D_0, since $e_1 \subseteq \bar{D}_0$; so $D_0 \subseteq \mathrm{ins}(C)$, as D_0 is connected and misses $U(C)$, and then $\mathrm{ins}(C)$ contains a point of F not in $U(C)$, which is impossible for a closed disc in Δ bounded by $U(C)$. Thus $v_0 \in V(W^*)$, and in the same way $v_{\theta/2} \in V(W^*)$. A closed walk through a vertex of level 0 and a vertex of level θ has length at least 2θ.[F11] In particular, since $\mathrm{ins}(W) \subseteq \Delta$, there is no such W^* with $U(W^*) \subseteq \mathrm{ins}(W)$. By (2) and [5, Theorem (8.13)], $d(e_1, e_{\theta/2}) \ge \theta$, as required. □

[p. 259] 7. Deletion

Our final objective is to investigate the effect on a respectful tangle in a drawing Γ, and on the associated metric, of deleting a part of Γ of small radius. We begin with the following.

[F11] **Proof gap.** The paper gives no proof of this sentence. The proof inserted here uses a level function on the vertices of K in Δ.

(7.1) *Let $\mathscr{T}$ be a tangle of order $\theta \geq 1$ in a graph G, and let $(A', B') \in \mathscr{T}$ have order* 1. *Let*

$$\mathscr{T}' = \{(A \cap B', B \cap B') : (A, B) \in \mathscr{T}\};$$

then $\mathscr{T}'$ is a tangle in B' of order θ.

Proof. Let $V(A' \cap B') = \{v\}$. Certainly every member of $\mathscr{T}'$ is a separation of B' of order $< \theta$, and we shall verify the three axioms. For the first axiom, let (A, B) be a separation of B' of order $< \theta$; we must show that one of (A, B), $(B, A) \in \mathscr{T}'$. Since $V(A \cup B) = V(B')$, we may assume that $v \in V(A)$ without loss of generality; and so $(A \cup A', B)$ is a separation of G of order $< \theta$. Hence $\mathscr{T}$ contains one of $(A \cup A', B)$, $(B, A \cup A')$, and so $\mathscr{T}'$ contains one of

$$((A \cup A') \cup B', B \cap B'), \quad (B \cap B', (A \cup A') \cap B'),$$

that is, one of (A, B), (B, A) as required. For the second axiom let $(A_i, B_i) \in \mathscr{T}$ $(1 \leq i \leq 3)$, so that $(A_i \cap B', B_i \cap B') \in \mathscr{T}$ for each i, and suppose that

$$(A_1 \cap B') \cup (A_2 \cap B') \cup (A_3 \cap B') = B'.$$

Thus $B' \subseteq A_1 \cup A_2 \cup A_3$. From the symmetry we may assume that $v \in V(A_1)$, and so $(A' \cup A_1, B' \cap B_1)$ has order at most that of (A_1, B_1). But $(A', B') \in \mathscr{T}$ and $(A_1, B_1) \in \mathscr{T}$, and so $(B' \cap B_1, A' \cup A_1) \notin \mathscr{T}$ by the second axiom since $A' \cup A_1 \cup (B' \cap B_1) = G$. Hence, $(A' \cup A_1, B' \cap B_1) \in \mathscr{T}$. But (A_2, B_2), $(A_3, B_3) \in \mathscr{T}$, and $(A' \cup A_1) \cup A_2 \cup A_3 = G$, contrary to the second axiom. This shows that $\mathscr{T}'$ satisfies the second axiom.

For the third, let $(A, B) \in \mathscr{T}$ and suppose that $V(A \cap B') = V(B')$, that is, $V(B') \subseteq V(A)$. By [4, Theorem (2.9)], $(B, A) \in \mathscr{T}$, a contradiction. Thus the third axiom holds. □

(7.2) *Let $\mathscr{T}$ be a respectful tangle of order $\theta \geq 2$ in a 2-cell drawing Γ in a connected surface Σ, and let its metric be d. Let F be a Γ-normal O-arc in Σ with $|F \cap V(\Gamma)| = 1$; let $F \cap V(\Gamma) = \{v\}$, let r be the region of Γ including $F - \{v\}$, let* $\mathrm{ins}(F) = \Delta$, *and let $\Gamma' = \Gamma|\overline{\Sigma - \Delta}$. Then Γ' is 2-cell, and $d(a, \{v\}) \leq 1$ for all $a \in A(\Gamma) - A(\Gamma')$. Moreover, let*

$$\mathscr{T}' = \{(A \cap \Gamma', B \cap \Gamma') : (A, B) \in \mathscr{T}\};$$

[p. 260] *then $\mathscr{T}'$ is a respectful tangle in Γ' of order θ. Let its metric be d', and let r' be the region of Γ' including r; then*

(i) *$d'(a, b) \geq d(a, b)$ for all $a, b \in A(\Gamma) \cap A(\Gamma')$, and*

(ii) $d'(a,r') \geq d(a,r)$ *for all* $a \in A(\Gamma) \cap A(\Gamma')$.

Proof. $r' = r \cup (\Delta - \{v\})$, and therefore r' is an open disc, by [5, Theorem (7.9)], since $\mathrm{bd}(\Delta) \cap r$ is connected. But every other atom of Γ' is an atom of Γ, and so Γ' is 2-cell. Clearly $d(a,\{v\}) \leq 1$ for every $a \in A(\Gamma) - A(\Gamma')$, since for any such a, either $a = r$ or $a \subseteq \mathrm{ins}(F)$. By (7.1), $\mathscr{T}'$ is a tangle in Γ' of order θ; we must show that it is respectful. Let F' be a Γ'-normal O-arc with $|F' \cap V(\Gamma')| < \theta$. We may assume, by "diverting" F' within r', that $F' \cap \mathrm{ins}(F) \subseteq \{v\}$, and so F' is Γ-normal and $|F' \cap V(\Gamma)| < \theta$. Let $\Delta' = \mathrm{ins}(F')$; then

$$(\Gamma \cap \Delta', \Gamma \cap \overline{\Sigma - \Delta'}) \in \mathscr{T},$$

and so

$$(\Gamma' \cap \Delta', \Gamma' \cap \overline{\Sigma - \Delta'}) \in \mathscr{T}'$$

by definition of $\mathscr{T}'$. Hence $\mathscr{T}'$ is respectful. Let ins' be its inside function, and d' its metric.

Let K be a radial drawing of Γ. Since r is incident with v "twice," there correspond two parallel edges e, f of K with ends u, v, say, so that $\Gamma \cap \Delta = \Gamma \cap \mathrm{ins}(C_0)$, where C_0 is the circuit of K with $E(C_0) = \{e, f\}$. We may therefore choose a radial drawing K' for Γ' so that K' is a subdrawing of K, by deleting from K f and every vertex and edge of K in $\mathrm{ins}(C_0) - U(C_0)$. Now $\mathrm{ins}'(C') = \mathrm{ins}(C')$ for every circuit C' of K' with $< 2\theta$ edges, because if $\Delta' = \mathrm{ins}(C')$, then as we saw above,

$$(\Gamma' \cap \Delta', \Gamma' \cap \overline{\Sigma - \Delta'}) \in \mathscr{T}'$$

and so $\Delta' = \mathrm{ins}'(C')$. Consequently $\mathrm{ins}(W) = \mathrm{ins}'(W)$ for every closed walk W of K' with length $< 2\theta$. To prove (i), let $a, b \in A(\Gamma) \cap A(\Gamma')$. To prove that $d'(a,b) \geq d(a,b)$, we may assume that $d'(a,b) < \theta$, since $d(a,b) \leq \theta$, and that $a \neq b$. Consequently there is a closed walk W of K' with length $2d'(a,b)$ such that $\mathrm{ins}'(W) = \mathrm{ins}(W)$ meets both a and b; hence, $d(a,b) \leq d'(a,b)$. Thus (i) holds. For (ii), let $a \in A(\Gamma) \cap A(\Gamma')$. To prove that $d'(a,r') \geq d(a,r)$ we may assume that $d'(a,r') < \theta$ and $a \neq r'$, and so there is a closed walk W of K' of length $2d'(a,r')$ such that $\mathrm{ins}'(W) = \mathrm{ins}(W)$ meets both a and r'. Consequently, $u \in \mathrm{ins}(W)$ (we recall that e, f have ends u, v) and so $\mathrm{ins}(W)$ meets a and r, and $d(a,r) \leq d'(a,r')$, and (ii) holds. □

[p. 261] (7.3) *Let* $\mathscr{T}$ *be a respectful tangle of order* $\theta \geq 2$ *in a 2-cell drawing* Γ *in a connected surface* Σ*, and let its metric be* d*. Let* $v \in V(\Gamma)$*, such that* $\Gamma \setminus v$ *is 2-cell. Then there is a respectful tangle* $\mathscr{T}'$ *of order* $\theta - 1$ *in* $\Gamma \setminus v$*, with metric* d'*, such that if* r' *is the region of* $\Gamma \setminus v$ *containing* v*, then*

(i) $d'(a,b) \geq d(a,b) - 2$ *for all* $a, b \in A(\Gamma) \cap A(\Gamma \setminus v)$*, and*

(ii) $d'(a, r') \geq d(a, \{v\}) - 1$ *for all* $a \in A(\Gamma) \cap A(\Gamma \setminus v)$.

Proof. Let $\Gamma' = \Gamma \setminus v$, and let

$$\mathscr{T}' = \{(A \setminus v, B \setminus v) : (A, B) \in \mathscr{T}, v \in V(A \cap B)\}.$$

By [4, Theorem (8.5)], $\mathscr{T}'$ is a tangle of order $\theta - 1$ in Γ'. Let us prove that it is respectful. We may therefore assume that Σ is not a sphere, for every tangle in a drawing in a sphere is respectful. Hence Γ is θ-representative, and so Γ' is $(\theta-1)$-representative. Let F be a Γ'-normal O-arc with $|F \cap V(\Gamma')| < \theta - 1$. Since Γ' is $(\theta-1)$-representative, there is a closed disc $\Delta \subseteq \Sigma$ bounded by F. Let $A' = \Gamma' \cap \Delta$, $B' = \Gamma' \cap \overline{\Sigma - \Delta}$; then (A', B') is a separation of Γ' of order $< \theta - 1$. Hence one of (A', B'), (B', A') belongs to $\mathscr{T}'$, and we assume for a contradiction that $(B', A') \in \mathscr{T}'$. Therefore there exists $(A, B) \in \mathscr{T}$ with $v \in V(A \cap B)$ and $A \setminus v = B'$, $B \setminus v = A'$. Since $\mathscr{T}$ is respectful, by [5, Theorem 6.4)] there is a circuit C of B with $V(A \cap C) = \emptyset$, such that $U(C)$ is non-null-homotopic in Σ. Since $v \in V(A)$ and hence $v \notin V(C)$, it follows that $C \subseteq B \setminus v = A'$ and so $U(C) \subseteq U(A') \subseteq \Delta$, a contradiction since $U(C)$ is non-null-homotopic. Hence $(A', B') \in \mathscr{T}'$, and so $\mathscr{T}'$ is respectful. Let ins' be the corresponding "inside" function.

To prove (ii), let K be a radial drawing of Γ. Now no two edges of K incident with v are parallel, since $\Gamma \setminus v$ is 2-cell.[F12] Moreover, a radial drawing of Γ' can be obtained from K by contracting all edges of K incident with v, and then repeatedly deleting one edge from any two parallel edges that bound a region, until none remain. Consequently we may choose a radial drawing K' of Γ' with $K' \cap (\Sigma - r) = K \cap (\Sigma - r)$ and with $v \in V(K')$. Let the edges of K' incident with v be $e_1, \ldots, e_k$ in cyclic order, where e_i has ends v, v_i, say; then to each e_i there corresponds a 2-edge path P_i of K with ends v, v_i, where in the natural sense $P_1, \ldots, P_k$ are again in cyclic order and each edge of K incident with a neighbour of v but not incident with v belongs to exactly one of $P_1, \ldots, P_k$. For any closed walk W of K', we denote by $\widetilde{W}$ the closed walk of K obtained from W by replacing each occurrence of each e_i by a 2-edge walk in the appropriate sense along P_i. Then the length of $\widetilde{W}$ is $2t$ more than the length of W, where v occurs t times in W.

[F12] **Proof gap.** This is false when Γ has a loop at v that bounds a disc. For example, take K_4 in the sphere, put a new vertex v in the middle of one edge, and add at v a loop e that bounds an empty disc inside a region r_2 at v. Then $\Gamma \setminus v$ is 2-cell, but r_2 meets v at two corners, on the two sides of e, so K has two parallel edges between v and the vertex of K in r_2. The proof can be completed by first applying (7.2) repeatedly, to remove the inside of every Γ-normal O-arc through v whose inside contains an edge of Γ. This removes such loops and does not decrease the metric. The rest of the argument is then as printed, and (7.3) holds as printed.

(1) *If v occurs t times in W and W has length $< 2\theta - 2t$, then* $\mathrm{ins}'(W) \subseteq \mathrm{ins}(\widetilde{W}) \cup r$.

[p. 262] *Subproof.* Certainly $U(W) \subseteq U(\widetilde{W}) \cup r$. Thus, let C' be a circuit of K' with $U(C') \subseteq U(W)$; we must show that $\mathrm{ins}'(C') \subseteq \mathrm{ins}(\widetilde{W}) \cup r$. Now C' has length $< 2\theta - 2t$. If $v \notin V(C)$ then C' is a circuit of K and $\mathrm{ins}'(C') = \mathrm{ins}(C')$ as we already saw, and $C' \subseteq K|\widetilde{W}$; and so $\mathrm{ins}(C') \subseteq \mathrm{ins}(\widetilde{W})$. We assume then that $v \in V(C')$. Let e_i, e_j be the two edges of C' incident with v. Let P be the path in $P_i \cup P_j$ between v_i and v_j; thus $|E(P)| = 2$ or 4, depending whether the middle vertices of P_i and P_j are equal or not. Let C be the circuit $P \cup (C' \setminus v)$; then $|E(C)| = |E(C')| + 2\varepsilon$, where $\varepsilon = 0$ if $|E(P)| = 2$ and $\varepsilon = 1$ if $|E(P)| = 4$. Let $\Delta = \mathrm{ins}(C)$, $A = \Gamma|\Delta$, $B = \Gamma|\overline{\Sigma - \Delta}$. Then $(A, B) \in \mathscr{T}$, and has order $\frac{1}{2}|E(C)|$. Let $A^+ \subseteq \Gamma$ be the drawing with vertex set $V(A) \cup \{v\}$ and edge set $E(A)$, and define B^+ similarly; then (A^+, B^+) has order $\leq \frac{1}{2}|E(C)| + 1 - \varepsilon \leq \frac{1}{2}|E(C')| + 1$. But $|E(C')| < 2\theta - 2t \leq 2\theta - 2$, and so (A^+, B^+) has order $< \theta$. By [4, Theorem (2.9)], $(A^+, B^+) \in \mathscr{T}$, and so $(A^+ \setminus v, B^+ \setminus v) \in \mathscr{T}'$; that is, $(\Gamma' \cap \Delta, \Gamma' \cap \overline{\Sigma - \Delta}) \in \mathscr{T}'$. Consequently, $\mathrm{ins}'(C') \subseteq \mathrm{ins}(C) \cup r \subseteq \mathrm{ins}(\widetilde{W}) \cup r$. This proves (1).

Now let us prove (i). Let $a, b \in A(\Gamma) \cap A(\Gamma')$. To show that $d'(a, b) \geq d(a, b) - 2$, we may assume that $d'(a, b) < \theta - 2$, since $d(a, b) \leq \theta$, and that $a \neq b$. Consequently there is a closed walk W of K' of length $2d'(a, b)$, such that a and b both meet $\mathrm{ins}'(W)$. By [5, Theorem (9.2)] W can be chosen so that v occurs at most twice in it. Hence $\widetilde{W}$ has length $\leq 2d'(a, b) + 4 < 2\theta$ and by (1), $\mathrm{ins}'(W) \subseteq \mathrm{ins}(\widetilde{W}) \cup r$. Since a, b meet $\mathrm{ins}'(W)$ and do not meet r, they both meet $\mathrm{ins}(\widetilde{W})$; and so $d(a, b) \leq d'(a, b) + 2$. Hence (i) holds.

To prove (ii), let $a \in A(\Gamma) \cap A(\Gamma')$; we must show that $d'(a, r) \geq d(a, \{v\}) - 1$. We may assume that $d'(a, r') < \theta - 1$, and so W exists as above (with $b = r$). But now W can be chosen so that v occurs at most once in W, since $v \in r$. Hence $\widetilde{W}$ has length $\leq 2d'(a, b) + 2$; and a and $\{v\}$ both meet $\mathrm{ins}(\widetilde{W})$, and so $d(a, v) \leq d'(a, r') + 1$. Hence (ii) holds. □

By combining (7.2) and (7.3), we deduce

(7.4) *Let $\mathscr{T}$ be a respectful tangle of order $\theta \geq 2$ in a 2-cell drawing Γ in a connected surface Σ, and let its metric be d. Let $v \in V(\Gamma)$. Then there is a 2-cell component Γ' of $\Gamma \setminus v$ and a respectful tangle $\mathscr{T}'$ of order $\theta - 1$ in Γ', with metric d', such that if r' is the region of Γ' containing v, then*

(i) $U(\Gamma) \subseteq U(\Gamma') \cup r'$

(ii) $d'(a, b) \geq d(a, b) - 2$ *for all* $a, b \in A(\Gamma) \cap A(\Gamma')$

(iii) $d'(a, r') \geq d(a, \{v\}) - 1$ *for all* $a \in A(\Gamma) \cap A(\Gamma')$, *and*

(iv) $d(a, \{v\}) \leq 2$ *for all* $a \in A(\Gamma)$ *with* $a \subseteq r'$.

Proof. We proceed by induction on $|E(\Gamma)|$. If there is no Γ-normal O-arc F with $F \cap V(\Gamma) = \{v\}$ such that ins(F) includes an edge of Γ, then $\Gamma \setminus v$ is 2-cell and by (7.3) we may take $\Gamma' = \Gamma \setminus v$ (because statements (i) and

[p. 263] (iv) of the theorem are then obviously true). We assume then that there exists F as in (7.2), where $F \subseteq \{v\} \cup r$ for some region r of Γ, and ins(F) includes an edge of Γ. Let Γ'' be obtained from Γ by deleting all vertices and edges in ins$(F) - F$, and let $\mathscr{T}'' = \{(A \cap \Gamma'', B \cap \Gamma'') : (A, B) \in \mathscr{T}\}$. By (7.2), Γ'' is 2-cell, and $\mathscr{T}''$ is a respectful tangle of order θ in Γ'', with metric d'' say, such that

(v) for all $a, b \in A(\Gamma) \cap A(\Gamma'')$, $d''(a, b) \geq d(a, b)$

(vi) for all $a \in A(\Gamma) - A(\Gamma'')$, $d(a, \{v\}) \leq 1$.

From the inductive hypothesis, there is a 2-cell component Γ' of $\Gamma'' \setminus v$ and, hence, of $\Gamma \setminus v$, and a respectful tangle $\mathscr{T}'$ of order $\theta - 1$ in Γ', with metric d', such that if r' is the region of Γ' containing v, then

(i') $U(\Gamma'') \subseteq U(\Gamma') \cup r'$

(ii') $d'(a, b) \geq d''(a, b) - 2$ for all $a, b \in A(\Gamma'') \cap A(\Gamma')$

(iii') $d'(a, r') \geq d''(a, \{v\}) - 1$ for all $a \in A(\Gamma'') \cap A(\Gamma')$, and

(iv') $d''(a, \{v\}) \leq 2$ for all $a \in A(\Gamma'')$ with $a \subseteq r'$.

We claim that Γ', $\mathscr{T}'$, r' satisfy the theorem. Certainly if r'' is the region of Γ'' including r, then $r'' \subseteq r'$ and hence by (i'),

$$U(\Gamma) \subseteq U(\Gamma'') \cup r'' \subseteq U(\Gamma') \cup r'$$

and (i) holds. For (ii), let $a, b \in A(\Gamma) \cap A(\Gamma')$; then $a, b \in A(\Gamma'')$, and by (ii') and (v),

$$d'(a, b) \geq d''(a, b) - 2 \geq d(a, b) - 2$$

and (ii) holds. For (iii), let $a \in A(\Gamma) \cap A(\Gamma')$; then $a \in A(\Gamma'')$, and by (iii) and (v),

$$d'(a, r') \geq d''(a, \{v\}) - 1 \geq d(a, \{v\}) - 1$$

and (iii) holds. For (iv), let $a \in A(\Gamma)$ with $a \subseteq r'$. If $a \notin A(\Gamma'')$ then by (vi), $d(a, \{v\}) \leq 1$, and (iv) holds. If $a \in A(\Gamma'')$ then by (v) and (iv'),

$$d(a, \{v\}) \leq d''(a, \{v\}) \leq 2,$$

and (iv) holds. □

(7.5) *Let* $\mathscr{T}$ *be a respectful tangle of order* $\theta \geq 3$ *in a 2-cell drawing* Γ *in a connected surface* Σ*, and let its metric be* d*. Let* r *be a region of* Γ*, and let* Z *be the set of vertices of* Γ *incident with* r*. Then there is a 2-cell*

[p. 264] subdrawing Γ' of $\Gamma \setminus Z$, and a respectful tangle $\mathscr{T}'$ of order $\theta - 2$ in Γ', with metric d', such that if r' is the region of Γ' with $r \subseteq r'$, then

(i) $U(\Gamma) \subseteq U(\Gamma') \cup r'$
(ii) $d'(a,b) \geq d(a,b) - 4$ for all $a, b \in A(\Gamma) \cap A(\Gamma')$
(iii) $d'(a,r') \geq d(a,r) - 2$ for all $a \in A(\Gamma) \cap A(\Gamma')$, and
(iv) $d(a,r) \leq 3$ for all $a \in A(\Gamma)$ with $a \subseteq r'$.

Proof. We proceed by induction on $|E(\Gamma)|$. First, suppose that there is a Γ-normal O-arc $F \subseteq \Sigma$ with $F \cap V(\Gamma) = \{v\}$ for some $v \in V(\Gamma)$, and with $F \subseteq r \cup \{v\}$, such that $ins(F)$ includes an edge of Γ. Let $\Delta = ins(F)$, let $\Gamma'' = \Gamma|\overline{\Sigma - \Delta}$, and let r'' be the region of Γ'' with $r \subseteq r''$. By (7.2), Γ'' is 2-cell and there is a respectful tangle $\mathscr{T}''$ of order θ in Γ_1, with metric d'', such that

(v) $d''(a,b) \geq d(a,b)$ for all $a, b \in A(\Gamma) \cap A(\Gamma'')$, and
(vi) $d''(a,r'') \geq d(a,r)$ for all $a \in A(\Gamma) \cap A(\Gamma'')$.

Let Z'' be the set of all vertices of Γ'' incident with r''. From the inductive hypothesis, there is a 2-cell subdrawing Γ' of $\Gamma'' \setminus Z''$, and a respectful tangle $\mathscr{T}'$ of order $\theta - 2$ in Γ', with metric d', such that if r' is the region of Γ' with $r'' \subseteq r'$, then

(i′) $U(\Gamma'') \subseteq U(\Gamma') \cup r'$
(ii′) $d'(a,b) \geq d''(a,b) - 4$ for all $a, b \in A(\Gamma'') \cap A(\Gamma')$
(iii′) $d'(a,r') \geq d''(a,r'') - 2$ for all $a \in A(\Gamma'') \cap A(\Gamma')$, and
(iv′) $d''(a,r'') \leq 3$ for all $a \in A(\Gamma'')$ with $a \subseteq r'$.

We claim that Γ', $\mathscr{T}'$, r' satisfy the theorem. Certainly Γ' is a subdrawing of $\Gamma \setminus Z$, since $Z \cap V(\Gamma'') \subseteq Z''$; and $r \subseteq r'' \subseteq r'$. To verify (i), by (i′),

$$U(\Gamma) \subseteq U(\Gamma'') \cup ins(F) \subseteq U(\Gamma'') \cup r'' \subseteq U(\Gamma') \cup r'.$$

Hence (i) holds. For (ii), let $a, b \in A(\Gamma) \cap A(\Gamma')$; then $a, b \in A(\Gamma'')$, and by (ii′) and (v),

$$d'(a,b) \geq d''(a,b) - 4 \geq d(a,b) - 4.$$

Hence (ii) holds. For (iii), let $a \in A(\Gamma) \cap A(\Gamma')$; then $a \in A(\Gamma'')$, and by (iii′) and (vi),

$$d'(a,r') \geq d''(a,r'') - 2 \geq d(a,r) - 2.$$

Hence (iii) holds. For (iv), let $a \in A(\Gamma)$ with $a \subseteq r'$. If $a \notin A(\Gamma'')$, then $a \cap ins(F) \neq \emptyset$, and so $d(a,r) \leq 1 \leq 3$. If $a \in A(\Gamma'')$, then by (vi) and (iv$'$), $d(a,r) \leq d''(a,r'') \leq 3$. Hence in either case (iv) holds.

[p. 265] We may therefore assume that there is no such O-arc F, and so $\bar{r} - r$ is an O-arc. Let Γ_1 be a drawing in Σ dual to Γ, in the usual sense of surface duals. For each $a \in A(\Gamma)$ there corresponds an atom of Γ_1 in the natural way, which we denote by a^*. Now Γ_1 is 2-cell, and has a common radial drawing with Γ. By [5, Theorem (6.6)], there is a respectful tangle $\mathscr{T}_1$ of order θ in Γ_1, with metric d_1, such that $d_1(a^*, b^*) = d(a,b)$ for all $a, b \in A(\Gamma)$. Since $\bar{r} - r$ is an O-arc, it follows that $\Gamma_2 = \Gamma_1 \setminus r^*$ is 2-cell. By (7.3) applied to Γ_1, there is a respectful tangle $\mathscr{T}_2$ of order $\theta - 1$ in Γ_2, with metric d_2, such that, if r_2 is the region of Γ_2 containing r^*, then

(v) $d_2(a,b) \geq d_1(a,b) - 2$ for all $a, b \in A(\Gamma_1) \cap A(\Gamma_2)$, and
(vi) $d_2(a, r_2) \geq d_1(a, r^*) - 1$ for all $a \in A(\Gamma_1) \cap A(\Gamma_2)$.

Let Γ_3 be a drawing in Σ dual to Γ_2—roughly, Γ_3 is obtained from Γ by contracting all edges of Γ included in $\bar{r} - r$. For each atom a of Γ_2 we denote by a^* the corresponding atom of Γ_3. Thus, r_2^* is the vertex of Γ_3 formed by contracting all the edges of Γ in $\bar{r} - r$, and we may arrange that $r_2^* \in r$, and $\Gamma_2 \setminus r_2^* = \Gamma \setminus Z$. Let $\mathscr{T}_3$ be the respectful tangle of order $\theta - 1$ in Γ_3 dual to $\mathscr{T}_2$ in the sense of [5, Theorem (6.6)];[F13] thus, $d_3(a^*, b^*) = d_2(a,b)$ for all $a, b \in A(\Gamma_2)$, where d_3 is the metric of $\mathscr{T}_3$. Now let us apply (7.4) to $\Gamma_3, \mathscr{T}_3, r_2^*$. Since $\Gamma_3 \setminus r_2^* = \Gamma \setminus Z$, we deduce that there is a 2-cell subdrawing Γ' of $\Gamma \setminus Z$ and a respectful tangle $\mathscr{T}'$ of order $\theta - 2$ in Γ' with metric d', such that if r' is the region of Γ' containing r_2^*, then

(vii) $U(\Gamma_3) \subseteq U(\Gamma') \cup r'$
(viii) $d'(a,b) \geq d_3(a,b) - 2$ for all $a, b \in A(\Gamma_3) \cap A(\Gamma')$
(ix) $d'(a, r') \geq d_3(a, r_2^*) - 1$ for all $a \in A(\Gamma_3) \cap A(\Gamma')$, and
(x) $d_3(a, r_2^*) \leq 2$ for all $a \in A(\Gamma_3)$ with $a \subseteq r'$.

We claim that $\Gamma', \mathscr{T}', r'$ satisfy the theorem. Certainly r' is the region of Γ' including r, because $r_2^* \in r \cap r'$. Now $Z \subseteq r'$ and so every edge of Γ incident

[F13] **Proof gap.** A gap in the proof: when $\theta = 3$ this uses [5, Theorem (6.6)] for order 2, where it fails for a drawing with a loop (see the notes to (6.5) and (6.6) in Graph Minors XI), and Γ_3 may have loops. The step still holds here because $\mathscr{T}_2$ comes from the tangle $\mathscr{T}_1$ of order 3. A loop of Γ_3 with no small side would give three small sides of $\mathscr{T}_1$ that cover Γ_1, against the second tangle axiom. So the pretangle induced in Γ_3 is a tangle, and (7.5) holds as printed.

with a member of Z is included in r'; and by (vii),

$$U(\Gamma \setminus Z) \subseteq U(\Gamma_3) \subseteq U(\Gamma') \cup r'.$$

Consequently, $U(\Gamma) \subseteq U(\Gamma') \cup r'$, and so (i) holds. For (ii), let $a, b \in A(\Gamma) \cap A(\Gamma')$. Then $a \cap r' = \emptyset$, and so $a^* \in A(\Gamma_2)$, and $a = a^{**} \in A(\Gamma_3)$. Similarly $b^* \in A(\Gamma_2)$ and $b = b^{**} \in A(\Gamma_3)$. By (viii) and (v),

$$d'(a,b) \geq d_3(a,b) - 2 = d_2(a^*, b^*) - 2 \geq d_1(a^*, b^*) - 4 = d(a,b) - 4$$

and so (ii) holds. For (iii), let $a \in A(\Gamma) \cap A(\Gamma')$; then $a^* \in A(\Gamma_2)$ and $a = a^{**} \in A(\Gamma_3)$. By (ix) and (vi),

$$d'(a,r') \geq d_3(a, r_2^*) - 1 = d_2(a^*, r_2) - 1 \geq d_1(a^*, r^*) - 2 = d(a,r) - 2,$$

[p. 266] and so (iii) holds. For (iv), let $a \in A(\Gamma)$ with $a \subseteq r'$. Then $a^* \in A(\Gamma_1)$. If $a^* \notin A(\Gamma_2)$ then either $a^* = r^*$ or a^* is an edge or region of Γ_1 incident with r^*, and in either case

$$2 \geq d_1(a^*, r^*) = d(a,r)$$

and (iv) holds. We assume then that $a^* \in A(\Gamma_2)$, and hence $a^{**} \in A(\Gamma_3)$. Now $a \notin A(\Gamma')$ and so $a \subseteq r'$, and hence $a^{**} \subseteq r'$. By (vi) and (x),

$$d(a,r) = d_1(a^*, r^*) \leq d_2(a^*, r_2) + 1 = d_3(a^{**}, r_2^*) + 1 \leq 3,$$

and so (iv) holds. □

(7.6) *Let $k \geq 0$ be an integer, and let $\mathscr{T}$ be a respectful tangle of order $\theta \geq 2k+1$ in a 2-cell drawing Γ in a connected surface Σ. Let its metric be d. Let r be a region of Γ, and let Z be the set of all vertices v of Γ such that there is a Γ-normal line F with $v \in F$, $F \cap r \neq \emptyset$ and $|F \cap V(\Gamma)| \leq k$. Then there is a 2-cell subdrawing Γ' of $\Gamma \setminus Z$, and a respectful tangle $\mathscr{T}'$ of order $\theta - 2k$ in Γ', with metric d', such that if r' is the region of Γ' with $r \subseteq r'$, then*

- (i) $U(\Gamma) \subseteq U(\Gamma') \cup r'$
- (ii) $d'(a,b) \geq d(a,b) - 4k$ for all $a, b \in A(\Gamma) \cap A(\Gamma')$
- (iii) $d'(a,r') \geq d(a,r) - 2k$ for all $a \in A(\Gamma) \cap A(\Gamma')$, and
- (iv) $d(a,r) \leq 2k+1$ for all $a \in A(\Gamma)$ with $a \subseteq r'$.

In particular, if $a, b \in A(\Gamma)$ and $a', b' \in A(\Gamma')$ with $a \subseteq a'$ and $b \subseteq b'$, then $d'(a', b') \geq d(a,b) - 4k - 2$.

Proof. Let us show first that the last claim of the theorem follows from (i)-(iv). Let $\Gamma', \mathscr{T}'$ satisfy (i)-(iv), and let $a, b \in A(\Gamma)$ and $a', b' \in A(\Gamma')$ with $a \subseteq a'$ and $b \subseteq b'$. If $a = a'$ and $b = b'$, then $a, b \in A(\Gamma')$ and the claim follows from (ii). We assume then that $b \neq b'$, and so $b' = r'$ from (i). If $a = a'$ then from (iii),

$$d'(a, r') \geq d(a, r) - 2k;$$

but from (iv), $d(b, r) \leq 2k + 1$, and so

$$d'(a, r') \geq d(a, r) + d(b, r) - 4k - 1 \geq d(a, b) - 4k - 1$$

as required. Finally, if $a \neq a'$, then $a' = b' = r'$ and from (iv),

$$d(a, b) \leq d(a, r) + d(b, r) \leq 2(2k + 1) = d'(a', b') + 4k + 2,$$

as required. Thus, it remains to find Γ' and $\mathscr{T}'$ satisfying (i)-(iv).

[p. 267] We proceed by induction on k. If $k = 0$ the result is trivial, taking $\Gamma' = \Gamma$ and $\mathscr{T}' = \mathscr{T}$, and so we assume that $k \geq 1$. Let Y be the set of vertices of Γ incident with r. By (7.5) there is a 2-cell component Γ'' of $\Gamma \setminus Y$, and a respectful tangle $\mathscr{T}''$ of order $\theta - 2$ in Γ'', with metric d'', such that if r'' is the region of Γ'' including r, then

- (v) $U(\Gamma) \subseteq U(\Gamma'') \cup r''$
- (vi) $d''(a, b) \geq d(a, b) - 4$ for all $a, b \in A(\Gamma) \cap A(\Gamma'')$
- (vii) $d''(a, r'') \geq d(a, r) - 2$ for all $a \in A(\Gamma) \cap A(\Gamma'')$, and
- (viii) $d(a, r) \leq 3$ for all $a \in A(\Gamma) - A(\Gamma'')$.

Let Z'' be the set of all vertices v of Γ'' such that there is a Γ''-normal line F with $v \in F$, $F \cap r'' \neq \emptyset$ and $|F \cap V(\Gamma'')| \leq k - 1$. Then $Z \subseteq Y \cup Z''$. From the inductive hypothesis (with $\mathscr{T}, \theta, k, \Gamma, r, Z$ replaced by $\mathscr{T}'', \theta - 2, k - 1, \Gamma'', r'', Z''$) there is a 2-cell subdrawing Γ' of $\Gamma'' \setminus Z''$ and a respectful tangle $\mathscr{T}'$ of order $(\theta - 2) - 2(k - 1)$ in Γ' with metric d', such that if r' is the region of Γ' with $r'' \subseteq r'$, then

- (i′) $U(\Gamma'') \subseteq U(\Gamma') \cup r'$
- (ii′) $d'(a, b) \geq d''(a, b) - 4(k - 1)$ for all $a, b \in A(\Gamma'') \cap A(\Gamma')$
- (iii′) $d'(a, r') \geq d''(a, r'') - 2(k - 1)$ for all $a \in A(\Gamma'') \cap A(\Gamma')$, and
- (iv′) $d''(a, r'') \leq 2(k - 1) + 1$ for all $a \in A(\Gamma'')$ with $a \subseteq r'$.

We claim that $\Gamma', \mathscr{T}', r'$ satisfy the theorem. For Γ' is a subdrawing of $\Gamma \setminus Z$, since $Z \subseteq Y \cup Z''$ and $Y \cap V(\Gamma') = \emptyset$; and r' is the region of Γ' including r'' and hence r. Now from (v) and (i′),

$$U(\Gamma) \subseteq U(\Gamma'') \cup r'' \subseteq U(\Gamma') \cup r',$$

and so (i) holds. For (ii), let $a, b \in A(\Gamma) \cap A(\Gamma')$; then $a, b \in A(\Gamma'')$, and from (ii′) and (vi),

$$d'(a,b) \geq d''(a,b) - 4(k-1) \geq d(a,b) - 4 - 4(k-1) = d(a,b) - 4k$$

and so (ii) holds. For (iii), let $a \in A(\Gamma) \cap A(\Gamma')$; then $a \in A(\Gamma'')$, and so from (iii′) and (vii),

$$d'(a,r') \geq d''(a,r'') - 2(k-1) \geq d(a,r) - 2k,$$

and so (iii) holds. For (iv), let $a \in A(\Gamma)$ with $a \subseteq r'$. If $a \notin A(\Gamma'')$ then from (viii), $d(a,r) \leq 3 \leq 2k+1$ and (iv) holds. If $a \in A(\Gamma'')$, then from (vii) and (iv′),

$$d(a,r) - 2 \leq d''(a,r'') \leq 2k - 1$$

and so (iv) holds. □

[p. 268] (7.7) *Let $\mathscr{T}$ be a respectful tangle of order θ in a 2-cell drawing Γ in a connected surface Σ; let its metric be d. Let Γ' be a 2-cell subdrawing of Γ, and let $\mathscr{T}'$ be a respectful tangle of order $\theta' \leq \theta$ in Γ', with metric d'. Then*

(i) *for $k \geq 0$, if $d'(a',b') \geq d(a,b) - k$ for all $a, b \in A(\Gamma)$ and $a', b' \in A(\Gamma')$ with $a \subseteq a'$ and $b \subseteq b'$, then $d(a,b) \geq \min(d'(a',b'), \theta - k)$ for all such a, b, a', b'*

(ii) *if $\theta = \theta'$ and $\mathscr{T}$ is the tangle in Γ induced by $\mathscr{T}'$, then $d(a,b) \geq d'(a',b')$ for all $a, b \in A(\Gamma)$ and $a', b' \in A(\Gamma')$ with $a \subseteq a'$ and $b \subseteq b'$.*

Proof. We prove both statements simultaneously. In case (ii), let $k = 0$. We assume that $a, b \in A(\Gamma)$, $a', b' \in A(\Gamma')$, $d(a,b) < d'(a',b') < \theta'$, and $d(a,b) < \theta - k$, and we shall show that neither case (i) nor case (ii) applies. Let K be a radial drawing of Γ, and let $\mathrm{ins}, \mathrm{ins}'$ be the inside functions of $\mathscr{T}$ and $\mathscr{T}'$. Now $a \neq b$, since $d'(a',b') > 0$ and, hence, $a' \neq b'$. But $d(a,b) < \theta$, and so there is a closed walk W of K of length $2d(a,b)$, such that $a \cap \mathrm{ins}(W)$, $b \cap \mathrm{ins}(W)$ are both non-empty. Let $\phi\colon S^1 \to \Sigma$ "follow" W in the natural sense; then $|\phi|$ is Γ-normal and has length $d(a,b)$ with respect to Γ, and $\mathrm{ins}(\phi)$ meets both a and b. Consequently, $|\phi|$ is Γ'-normal and has length $\leq d(a,b) < \theta'$ with respect to Γ'. Since $d'(a',b') > d(a,b)$,

$\mathrm{ins}'(\phi)$ does not meet both a' and b' by (2.1), and, hence, we may assume that $a \cap \mathrm{ins}'(\phi) = \emptyset$, since $a \subseteq a'$. Consequently $a \cap |\phi| = \emptyset$, and since $a \cap \mathrm{ins}(\phi) \neq \emptyset$, there is an O-arc $F \subseteq |\phi|$ with $a \subseteq \mathrm{ins}(F)$. Now F is Γ-normal, and $|F \cap V(\Gamma)| \leq d(a,b) < \theta'$. Since $a \cap \mathrm{ins}'(F) = \emptyset$, it follows that $\mathrm{ins}(F) \neq \mathrm{ins}'(F)$, and so $\mathscr{T}$ is not induced by $\mathscr{T}'$. Hence the hypothesis of case (ii) does not hold. Choose $c \in A(\Gamma)$ such that $F \cap c \neq \emptyset$. By (1.2), there exists $f \in E(\Gamma)$ such that $d(c,f) = \theta$. Let $c', f' \in A(\Gamma')$ with $c \subseteq c'$ and $f \subseteq f'$. Then $f \notin \mathrm{ins}(F)$, since $c \cap \mathrm{ins}(F) \neq \emptyset$ and $|F \cap V(\Gamma)| < \theta$. Since $\mathrm{ins}(F) \neq \mathrm{ins}'(F)$, it follows that $f \subseteq \mathrm{ins}'(F)$, and so by (2.1),

$$d'(c', f') \leq |F \cap V(\Gamma')| \leq d(a,b) < \theta - k = d(c,f) - k.$$

Consequently, the hypothesis of case (i) does not hold, as required. □

We deduce

(7.8) *Let $k \geq 0$ be an integer, and let $\mathscr{T}$ be a respectful tangle of order $\theta \geq 2k+2$ in a 2-cell drawing Γ in a connected surface Σ. Let its metric be d. Let $c \in A(\Gamma)$, and let Γ' be a 2-cell subdrawing of Γ such that $d(c,e) \leq k$ for all $e \in E(\Gamma) - E(\Gamma')$, and such that if $c \in E(\Gamma') - E(\Gamma')$ then $k \geq 2$.*[F14] *Then there is a respectful tangle $\mathscr{T}'$ of order $\theta - 2k$ in Γ', with metric d', such that $d'(a',b') \geq d(a,b) - 4k - 2$ for all $a, b \in A(\Gamma)$ and $a', b' \in A(\Gamma')$ with $a \subseteq a'$ and $b \subseteq b'$.*

[p. 269] *Proof.* Let K be a radial drawing of Γ, and let h be the atom of K corresponding to c. Now either $h = \{v\}$ for some $v \in V(K)$, or $2k \geq 4$ and h is a region of K incident with ≤ 4 edges of K; and so in either case there is a closed walk W_0 of K of length $\leq 2k$ with $h \subseteq \mathrm{ins}(W_0)$, that is, with $c \cap \mathrm{ins}(W_0) \neq \emptyset$. Choose W_0 with $\mathrm{ins}(W_0)$ maximal.

If $V(W) \subseteq V(\Gamma)$ then $V(W) = \{v\}$ for some $v \in V(\Gamma)$; and hence $k = 0$ and $\Gamma' = \Gamma$, and the theorem is satisfied taking $\mathscr{T}' = \mathscr{T}$. We may assume, therefore, that there exists $v_0 \in V(W_0) - V(\Gamma)$. Let r be the region of Γ with $v_0 \in r$. Let Z be the set of all vertices v of Γ such that there is a Γ-normal line F with $v \in F, F \cap r \neq \emptyset$ and $|F \cap V(\Gamma)| \leq k$ (or equivalently, all $v \in V(\Gamma)$ such that there is a path of K between v and v_0 with $\leq 2k$ edges). By (7.6), there is a 2-cell subdrawing Γ'' of $\Gamma \setminus Z$, and a respectful tangle $\mathscr{T}''$ of order $\theta - 2k$ in Γ'', with metric d'', such that

$$d''(a'', b'') \geq d(a,b) - 4k - 2$$

[F14] **Misprint.** The condition meant is "if $c \in E(\Gamma) - E(\Gamma')$ then $k \geq 2$", as in (7.10). The proof uses $k \geq 2$ only when $c = e$ for some $e \in E(\Gamma) - E(\Gamma')$. As printed, (7.8) fails when $c \in E(\Gamma) - E(\Gamma')$ and $k = 0$, and it does not give (7.10) when $c \in E(\Gamma')$ and $k \leq 1$. With the corrected condition, (7.8) holds.

for all $a, b \in A(\Gamma)$ and $a'', b'' \in A(\Gamma'')$ with $a \subseteq a''$ and $b \subseteq b''$, and such that $d(r, f) \leq 2k + 1$ for all $f \in E(\Gamma) - E(\Gamma'')$.

We claim that $\Gamma'' \subseteq \Gamma'$. For suppose not. Since Γ'' has no isolated vertices (because it is 2-cell and has a tangle of order ≥ 2) there is an edge e of Γ'' with $e \notin E(\Gamma')$. Since $e \in E(\Gamma) - E(\Gamma')$ it follows from the hypothesis that $d(c, e) \leq k$. Since $k < \theta$ (and since $k \geq 2$ if $e = c$) there is a closed walk W_1 of K of length $\leq 2k$ such that $e \subseteq \text{ins}(W_1)$ and $c \cap \text{ins}(W_1) \neq \emptyset$. Now $\text{ins}(W_0) \cup \text{ins}(W_1) \neq \Sigma$, by [5, Theorem (8.2)], and $\text{ins}(W_0) \not\subseteq \text{ins}(W_1)$ by the maximality of W_0. But $bd(\text{ins}(W_0))$ and $bd(\text{ins}(W_1))$ are both arc-wise connected, by [5, Theorem (4.4)], and so by [5, Theorem (7.7)], either $\text{ins}(W_1) \subseteq \text{ins}(W_0)$ or $V(W_1) \cap V(W_0) \neq \emptyset$. Since $e \subseteq \text{ins}(W_1)$, it follows in either case that there is a closed walk W in K of length $\leq 4k$ with $v_0 \in V(W)$ and $e \subseteq \text{ins}(W)$. Since W has length $\leq 4k$, it follows from the definition of Z that $V(W) \cap V(\Gamma) \subseteq Z$ and hence $V(W) \cap V(\Gamma'') = \emptyset$. Since Γ'' is connected, $U(\Gamma'') \cap bd(\text{ins}(W)) = \emptyset$, and $e \subseteq U(\Gamma'') \cap \text{ins}(W)$, it follows that $U(\Gamma'') \subseteq \text{ins}(W)$. Now by (1.2), there is an edge f of Γ such that $d(r, f) = \theta$. Since $\theta > 2k + 1$, it follows from the choise of Γ'' that $f \in E(\Gamma'')$, and hence $f \subseteq \text{ins}(W)$. Consequently, $d(r, f) \leq 2k < \theta$, a contradiction. This proves that $\Gamma'' \subseteq \Gamma'$.

Let $\mathscr{T}'$ be the tangle of order $\theta - 2k$ induced by $\mathscr{T}''$ in Γ'. Then $\mathscr{T}'$ is respectful; let its metric be d'. Let $a, b \in A(\Gamma)$ and $a', b' \in A(\Gamma')$ with $a \subseteq a'$ and $b \subseteq b'$; and let $a'', b'' \in A(\Gamma'')$ such that $a' \subseteq a''$ and $b' \subseteq b''$. By the choice of Γ'',

$$d''(a'', b'') \geq d(a, b) - 4k - 2.$$

But from (7.7)(ii), $d'(a', b') \geq d''(a'', b'')$, and so $\mathscr{T}'$ satisfies the theorem. □

[p. 270] (7.9) *Let $k \geq 0$ be an integer. Let $\mathscr{T}$ be a respectful tangle of order $\theta > k$ in a 2-cell drawing Γ in a connected surface Σ; let its metric be d. Let Γ' be a 2-cell subdrawing of Γ, and let $\mathscr{T}'$ be a respectful tangle of order $\theta - k$ in Γ', with metric d', such that $d'(a', b') \geq d(a, b) - k$ for all $a, b \in A(\Gamma)$ and $a', b' \in A(\Gamma')$ with $a \subseteq a'$ and $b \subseteq b'$. Then $(A \cap \Gamma', B \cap \Gamma') \in \mathscr{T}'$ for all $(A, B) \in \mathscr{T}$ of order $< \theta - k$. Moreover, if $\mathscr{T}''$ is another respectful tangle of order $\theta - k$ in Γ', with metric d'', and again $d''(a', b') \geq d(a, b) - k$ for all $a, b \in A(\Gamma)$ and $a', b' \in A(\Gamma')$ with $a \subseteq a'$ and $b \subseteq b'$, then $\mathscr{T}'' = \mathscr{T}'$.*

Proof. Let $\mathscr{T}_1$ be the tangle of order $\theta - k$ induced in Γ by $\mathscr{T}'$, and let $\mathscr{T}_2$ be the set of members of $\mathscr{T}$ of order $< \theta - k$. To show that $(A \cap \Gamma', B \cap \Gamma') \in \mathscr{T}'$ for all $(A, B) \in \mathscr{T}$ of order $< \theta - k$, we must show that $\mathscr{T}_1 = \mathscr{T}_2$. Certainly $\mathscr{T}_1$ and $\mathscr{T}_2$ are both respectful tangles of order $\theta - k$ in Γ; let their metrics be d_1 and d_2, and let their inside functions be $\text{ins}_1, \text{ins}_2$. Let K be a

radial drawing of Γ, and suppose that $\mathscr{T}_1 \neq \mathscr{T}_2$. Then by [5, Theorem (6.3)], there is a circuit C of K of length $< 2(\theta - k)$ such that $\mathrm{ins}_1(C) \neq \mathrm{ins}_2(C)$. Choose $v \in V(C)$. By (1.2) there is a region r of K such that $d(\{v\}, r) = \theta$. Hence $d_2(\{v\}, r) = \theta - k$ by definition of $\mathscr{T}_2$, and so $r \notin \mathrm{ins}_2(C)$; and $d_1(\{v\}, r) = \theta - k$, since $\{v\}$ and r both correspond to atoms of Γ', and so $r \notin \mathrm{ins}_1(C)$. This is impossible, since $\mathrm{ins}_1(C) \cup \mathrm{ins}_2(C) = \Sigma$. We deduce that $\mathscr{T}_1 = \mathscr{T}_2$, and the first claim of the theorem holds.

For the second, let $\mathscr{T}''$ be as described in the theorem. Let K' be a radial drawing of Γ' and suppose that $\mathscr{T}' \neq \mathscr{T}''$. Then by [5, Theorem (6.3)] there is a circuit C of K' with length $< \theta - k$ such that $\mathrm{ins}'(C) \neq \mathrm{ins}''(C)$. Let a be an atom of Γ with $a \cap U(C) \neq \emptyset$. By (1.2) there is a region r of Γ with $d(a, r) = \theta$. Let $a', b' \in A(\Gamma')$ with $a \subseteq a'$ and $b \subseteq b'$; then

$$d'(a', b'), d''(a', b') \geq d(a, r) - k = \theta - k$$

from the hypothesis. Since C has length $< \theta - k$ it follows that $b' \notin \mathrm{ins}'(C), \mathrm{ins}''(C)$, a contradiction. Thus $\mathscr{T}' = \mathscr{T}''$. $\square$

We see in (7.8) that, while the order of $\mathscr{T}'$ is only $2k$ less than that of $\mathscr{T}$, the discrepancy in the distance functions $d(a, b) - d'(a', b')$ is larger, up to $4k + 2$. No doubt this is not best possible; perhaps the right answer is $2k$. But in view of (7.7) and (7.9), there are advantages in having these two discrepancies equal, and we therefore shall discard $2k + 2$ from the order of $\mathscr{T}'$, to obtain the following.

(7.10) *Let $k \geq 0$ be an integer, and let $\mathscr{T}$ be a respectful tangle of order $\geq 4k+3$ in a 2-cell drawing Γ in a connected surface Σ. Let its metric be d. Let $c \in A(\Gamma)$, and let Γ' be a 2-cell subdrawing of Γ such that $d(c, e) \leq k$ for all $e \in E(\Gamma) - E(\Gamma')$, and such that if $c \in E(\Gamma) - E(\Gamma')$ then* [p. 271] *$k \geq 2$. Then there is a respectful tangle $\mathscr{T}'$ of order $\theta - 4k - 2$ in Γ', with metric d', such that $d'(a', b') \geq d(a, b) - 4k - 2$ for all $a, b \in A(\Gamma)$ and $a', b' \in A(\Gamma')$ with $a \subseteq a'$ and $b \subseteq b'$. Moreover,*

(i) *$\mathscr{T}'$ is unique*

(ii) *$(A \cap \Gamma', B \cap \Gamma') \in \mathscr{T}'$ for all $(A, B) \in \mathscr{T}$ of order $< \theta - 4k - 2$,*

(iii) *$d'(a', b') \leq d(a, b)$ for all $a, b \in A(\Gamma)$ and $a', b' \in A(\Gamma')$ with $a \subseteq a'$ and $b \subseteq b'$.*

Proof. From (7.8) there is a respectful tangle $\mathscr{T}''$ order $\theta - 2k$ in Γ', with metric d'', such that

$$d''(a', b') \geq d(a, b) - 4k - 2$$

for all $a, b \in A(\Gamma)$ and $a', b' \in A(\Gamma')$ with $a \subseteq a'$ and $b \subseteq b'$. Let $\mathscr{T}'$ be the set of all members of $\mathscr{T}''$ of order $< \theta - 4k - 2$. Then $\mathscr{T}'$ is respectful tangle

of order $\theta - 4k - 2$ in Γ', with metric d'; and for $a', b' \in A(\Gamma')$,

$$d'(a', b') = \min(d''(a', b'), \theta - 4k - 2).$$

Thus for $a, b \in A(\Gamma)$ and $a', b' \in A(\Gamma')$ with $a \subseteq a'$ and $b \subseteq b'$,

$$\begin{aligned} d'(a', b') &= \min(d''(a', b'), \theta - 4k - 2) \\ &\geq \min(d(a, b) - 4k - 2, \theta - 4k - 2) = d(a, b) - 4k - 2 \end{aligned}$$

and so $\mathscr{T}'$ satisfies the first claim of the theorem. Moreover, (i) and (ii) follows from (7.9). For (iii), let a, b, a', b' be as usual; then from (7.7)(i),

$$d(a, b) \geq \min(d'(a', b'), \theta - 4k - 2) = d'(a', b')$$

and so (iii) holds. □

The following result, complementary to (7.10), will also be useful.

(7.11) *Let Γ be a 2-cell drawing with $E(\Gamma) \neq \emptyset$, in a connected surface Σ, with a respectful tangle $\mathscr{T}$ of order θ, with metric d. Let $z \in A(\Gamma)$ and let $\lambda \geq 0$ be an integer with $\lambda < \theta$, such that if $z \in E(\Gamma)$ then $\lambda \geq 2$.*[F15] *Let Γ' be the subdrawing of Γ formed by all edges e of Γ with $d(z, e) > \lambda$ and their ends. Then Γ' is 2-cell.*

Proof. Let K be a radial drawing of Γ, and let c be the atom of K corresponding to z. Let Z be the union of $\mathrm{ins}(W)$, taken over all closed walks W of K of length $\leq 2\lambda$ with $c \subseteq \mathrm{ins}(W)$. By [5, Theorem (8.10)] and (1.2), Z is simply-connected and $Z \neq \Sigma$. From [5, Theorem (7.6)], it follows that $\Sigma - Z$ is non-null, open, and arc-wise connected. Consequently,

[p. 272] (1) *If $r, r' \in R(K)$ with $r, r' \subseteq \Sigma - Z$, then there is a sequence $r = r_1, r_2, \ldots, r_t = r'$ of regions of K, all included in $\Sigma - Z$, such that for $1 \leq i < t$ there is an edge of K incident with r_i and r_{i+1}.*

Now for each $e \in E(\Gamma)$ there is a region r of K with $e \subseteq r$; and $e \in E(\Gamma')$ if and only if $r \not\subseteq Z$, that is, $r \subseteq \Sigma - Z$. Conversely, every region of K in

[F15] **Statement mistake.** As printed, (7.11) fails when z is an edge and $\lambda \leq 1$. In the sphere, let Γ be one edge e_0, with $\theta = 2$, $z = e_0$ and $\lambda = 0$. No edge has $d(z, e) > 0$, so Γ' is empty and is not 2-cell. Asking for $|E(\Gamma)| \geq 2$ is not enough. Let Γ be a path with three edges f_1, z, f_2 in the sphere, and let $\mathscr{T}$ be the tangle of order 2 whose small sides are the sides that do not contain z. It is respectful, since every Γ-normal O-arc that meets at most one vertex bounds a disc on the side away from z. Let $\lambda = 1$. No closed walk of K of length less than 4 has z in its inside, so $d(z, f_1) = d(z, f_2) = 2 > \lambda$, while $d(z, z) = 0$. So Γ' consists of f_1, f_2 and their ends, and it is not connected, so it is not 2-cell. The proof fails at "$e \in E(\Gamma')$ if and only if $r \not\subseteq Z$": here $Z = \emptyset$, and yet $z \notin E(\Gamma')$. The hypothesis inserted here is the one that (7.8) and (7.10) have for c, and with it (7.11) holds. The proof uses (1.2) only to show that $Z \neq \Sigma$, and this also follows from [5, Theorem (8.12)], so no hypothesis on $|E(\Gamma)|$ is needed.

$\Sigma - Z$ includes an edge of Γ'. From (1), since $\Sigma - Z$ is open and non-null, we deduce that Γ' is non-null and connected. To show that Γ' is 2-cell, it therefore suffices to show that if $\phi\colon S^1 \to \Sigma$ is continuous and $|\phi| \cap U(\Gamma') = \emptyset$ then ϕ is null-homotopic in Σ. But it is easily seen that ϕ is homotopic in Σ to a continuous $\phi'\colon S^1 \to \Sigma$ with $|\phi'| \subseteq Z$; and since Z is simply-connected, ϕ' (and hence ϕ) is null-homotopic in Σ. The result follows. $\square$

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